

Project ARIA: A Cognitive Multi-Agent Vision-Language-Action Architecture for Autonomous Manipulation on Low-Cost 5-DoF Robotic Arms

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Abstract—Autonomous robotic manipulation has entered a transformative era driven by Vision-Language-Action (VLA) foundation models. However, contemporary robotic foundation policies are monolithic black boxes demanding datacenter-scale compute (> 24 GB VRAM), incurring prohibitive inference latencies (> 100 ms), and failing catastrophically on low-cost underactuated manipulators. In this paper, we present the research formulation, architectural design, physical implementation, and exhaustive empirical validation of Project ARIA (Autonomous Reasoning & Interaction Agent), a decentralized cognitive multi-agent VLA architecture specifically engineered to deliver high-precision physical autonomy on budget 5-DoF manipulators within an 8.0 GB VRAM budget.

ARIA addresses the foundational divide between high-level semantic cognition and low-level deterministic motor execution through a stratified six-tier hierarchy comprising fifteen specialized, lifecycle-managed software agents communicating asynchronously across a unified State Bus. To eliminate costly active depth cameras, we formulate a Zero-Cost-Depth perception pipeline combining overhead YOLOv8m detection and SAM2 zero-shot segmentation with wrist-mounted Depth-Anything v2 monocular metric depth recovery, grounded by the physical end-effector claw geometry ($Z_{\text{tips}} = 0.065$ m) to achieve sub-centimeter localization accuracy ($RMSE = 8.2$ mm). For deterministic execution, we derive an exact closed-form analytical Inverse Kinematics (IK) solver resolving the 5-DoF anthropomorphic chain in 0.08 ms ($231\times$ faster than numerical optimization) with 100% reachability determinism. Cognitive reasoning is orchestrated by an open-weight quantized LLM executing Tree-of-Thoughts task decomposition with an emoji-annotated Chain-of-Thought (CoT) telemetry stream and Human-in-the-Loop (HITL) confirmation gates.

Benchmarked across 10 standardized physical and simulated manipulation tasks, ARIA achieves an overall manipulation success rate of 89.0%, substantially outperforming state-of-the-art imitation learning and foundation policies including LeRobot ACT (72.0%), OpenVLA (68.0%), and Physical Intelligence π_0 (61.0%). Furthermore, in an automated manufacturing workcell featuring continuous moving conveyor lines

($v_{\text{belt}} = 0.05$ m/s), ARIA demonstrates predictive rendezvous grasping ($\Delta v \leq 0.002$ m/s) and dual-bin sorting into a 4-pocket palletizing tray ($RMSE = 1.14$ mm) and scrap reject bin, slashing complete system deployment costs by over 95%.

Index Terms—Cognitive robotics, multi-agent systems, vision-language-action (VLA) models, low-cost manipulators, zero-cost depth estimation, inverse kinematics, conveyor tracking, explainable artificial intelligence.

I. Introduction

THE realization of autonomous physical agents capable of executing complex manipulation in unstructured human environments represents a foundational ambition of robotics and embodied artificial intelligence [1]. Recent advances in dynamic trajectory planning, visual servoing, and predictive object interception established by Sun et al. [1] within ROS and Gazebo simulation frameworks have demonstrated that accurate kinematic forecasting is essential for grasping moving targets. However, traditional visual servoing frameworks rely on static, hand-engineered geometric primitives that fail to generalize across diverse operational contexts, highlighting the critical imperative for architectures capable of adaptive semantic reasoning and autonomous recovery.

To endow manipulators with generalizable semantic comprehension, the robotics community has increasingly embraced large-scale Vision-Language-Action (VLA) foundation models. Kim et al. [2] established OpenVLA, an open-source 7-billion-parameter foundation model fine-tuned on the massive Open X-Embodiment dataset across diverse robotic embodiments. While OpenVLA demonstrates remarkable semantic zero-shot generalization to novel visual objects, its autoregressive transformer architecture demands over 16 GB of dedicated VRAM and introduces a control loop inference latency of 140 ms per step. On physical robotic hardware, this computational lag introduces severe control instability, preventing responsive reaction to dynamic environmental events or unexpected mechanical slippage.

To mitigate multi-step planning brittleness, Zhao et al. [3] introduced CoT-VLA, an architecture that leverages visual chain-of-thought prompting to explicitly predict intermediate visual sub-goals and spatial bounding regions prior to generating continuous action tokens. Although

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CoT-VLA enhances multi-step spatial consistency, its dependency on monolithic autoregressive decoding compounds execution latency, requiring up to 250–400 ms per forward pass. Such latencies render monolithic chain-of-thought foundation models incompatible with real-time reactive trajectory control, where motor updates must execute at frequencies exceeding 50 Hz to ensure physical stability.

Alternative foundation formulations explore diffusion-based continuous trajectory generation. Ghosh et al. [4] proposed Octo, a generalist robotic policy trained on multi-task demonstration datasets using a diffusion transformer backbone supporting flexible action chunking and multi-camera inputs. While Octo achieves smooth trajectory synthesis and handles multi-modal action distributions effectively, its iterative denoising process requires significant GPU compute resources (> 12 GB VRAM) and operates as an uninterpretable latent black box, providing no explicit kinematic verification or collision guarantees prior to joint actuation.

At the frontier of dexterous physical manipulation, Black et al. [5] engineered π_0 (Physical Intelligence), a 3-billion-parameter flow-matching foundation model capable of executing high-frequency tactile and visual manipulation across diverse hardware configurations. Nevertheless, the physical deployment of π_0 presupposes enterprise-grade computing clusters equipped with high-end datacenter GPUs (e.g., NVIDIA H100/A100). This computational dependency creates an insurmountable deployment barrier for educational laboratories, small-scale industrial workcells, and consumer robotics, where compute is strictly restricted to consumer-grade edge hardware.

Furthermore, mainstream embodied foundation research exhibits a profound economic disconnect from accessible robotics hardware. Wu et al. [6] introduced GELLO, an intuitive low-cost teleoperation and imitation learning framework utilizing passive mechanical replicas to demonstrate that budget hardware can acquire complex manipulation skills. Yet, most downstream manipulation policies trained via imitation learning continue to mandate research-grade 7-DoF robotic arms (e.g., Franka Emika Panda, Kinova Gen3, UR5e) costing \$25,000–\$60,000. Conversely, budget 5-DoF articulated manipulators ($< \$150$) driven by PWM or serial bus servos remain largely neglected due to physical underactuation, joint elasticity, mechanical backlash, and an absence of joint torque sensing.

A parallel challenge lies in the sensory domain, where standard manipulation setups depend on active structured-light or Time-of-Flight (ToF) RGB-D sensors (e.g., Intel RealSense D435i, Photoneo). As documented by the ClearDepth Team [7], active infrared depth sensors suffer from severe minimum-range dead zones (< 0.2 m), multi-path reflections on metallic or specular surfaces, and total signal dropout on transparent glassware. In close-range desktop manipulation (0.05–0.25 m), where the end-effector approaches the workpiece, active depth cameras frequently fail, leaving the robot blind during the most

critical grasping phase.

To circumvent active sensor failure modes, monocular depth foundation models have advanced rapidly. Yang et al. [8] developed Depth-Anything v2, establishing unprecedented robustness in monocular relative depth estimation across diverse lighting and environmental conditions. Nevertheless, monocular vision transformer models output normalized relative disparity maps rather than physical metric distance. Without physical grounding or known geometric anchors, raw monocular depth predictions cannot be directly transformed into Cartesian coordinates for robot tool positioning, limiting their standalone utility in physical manipulation.

On the algorithmic control level, industrial manipulation frameworks (such as MoveIt2, KDL, and TRAC-IK) rely on numerical optimization routines utilizing damped least-squares (DLS) or Jacobian transpose inversions. As proved by Wagaa and Kallel [9], numerical Inverse Kinematics (IK) solvers exhibit severe algorithmic vulnerabilities on articulated chains lacking spherical wrists, incurring 10–20 ms calculation latencies, getting trapped in local minima, and diverging near kinematic singularities. On underactuated 5-DoF arms, rank deficiency in the 6×5 geometric Jacobian causes standard numerical solvers to fail on boundary targets, demanding exact closed-form algebraic formulations.

Finally, in dynamic manufacturing environments, manipulators must track, intercept, and sort items moving on continuous industrial conveyor belts. Sahu et al. [10] demonstrated autonomous visual object tracking on a 2-DOF robotic manipulator within Gazebo simulation, highlighting the critical role of predictive intercept calculation. However, translating dynamic conveyor interception to multi-DoF articulated arms while performing real-time quality defect inspection and feedforward velocity matching under edge compute constraints remains an open challenge.

A. Research Gaps Identified

Synthesizing the state of the art reveals five fundamental, unresolved research gaps:

- 1) Gap 1: The Compute-Edge Disconnect in VLA Deployments: Existing foundation models mandate datacenter-grade multi-GPU infrastructure (> 24 – 80 GB VRAM). No unified cognitive architecture exists that partitions foundation capabilities into modular cooperating agents executing deterministically on a single consumer laptop GPU (< 8.0 GB VRAM).
- 2) Gap 2: Black-Box Opacity vs. Industrial Causal Diagnostics: Monolithic policies map pixels directly to motor actions without exposing intermediate spatial reasoning, inverse kinematics health, or contact physics. Consequently, when manipulation fails, operators cannot diagnose the root cause or trigger automated recovery.

- 3) Gap 3: Close-Range Sensor Dead-Zones and Metric Scale Ambiguity: Active structured-light depth cameras fail below 0.2 m and scatter on specular surfaces, while state-of-the-art monocular depth models provide only unscaled relative disparity maps lacking metric scale for Cartesian trajectory generation.
- 4) Gap 4: Numerical IK Instability on Underactuated 5-DoF Chains: General-purpose numerical IK solvers diverge near kinematic singularities and incur 10–20 ms latencies on 5-DoF arms lacking independent wrist roll/yaw axes, creating a need for deterministic sub-millisecond analytical solutions.
- 5) Gap 5: Dynamic Conveyor Rendezvous Under Budget Constraints: Prior moving-target manipulation literature assumes high-torque industrial 6/7-DoF arms backed by optical PLC shaft encoders. Achieving dynamic moving conveyor tracking ($\mathbf{v}_{ee} \approx \mathbf{v}_{belt}$) and dual-bin sorting on budget servo-driven arms remains completely unexplored.

B. Contributions of This Work

To bridge these gaps, this article presents Project ARIA (Autonomous Reasoning & Interaction Agent), establishing the following core scientific and engineering contributions:

- 1) Decentralized 15-Agent Cognitive Architecture: We design, implement, and validate a non-monolithic multi-agent architecture where 15 specialized agents operate as ROS 2 LifecycleNode instances across six functional tiers, guaranteeing fault isolation, modularity, and sub-5 ms emergency preemption within an 8.0 GB VRAM budget.
- 2) Zero-Cost-Depth Perception Pipeline: We formulate a markerless perception pipeline fusing overhead eye-to-hand RGB video (YOLOv8m detection, SAM2 instance segmentation) with wrist-mounted eye-in-hand Depth-Anything v2 monocular depth, mathematically grounded via physical claw geometry ($Z_{tips} = 0.065$ m) to achieve sub-centimeter metric accuracy ($RMSE = 8.2$ mm) without active infrared depth sensors.
- 3) Sub-Millisecond Closed-Form Analytical Kinematics: We derive an exact algebraic closed-form Inverse Kinematics solver resolving the 2R planar subproblem for 5-DoF anthropomorphic arms in 0.08 ms ($231\times$ faster than numerical optimization) with 100% reachability determinism and zero singularity divergence.
- 4) Hierarchical LLM Planning with Explainable CoT Telemetry: We integrate local quantized LLMs (Mistral-7B / Llama 3.1 8B via Ollama) utilizing Tree-of-Thoughts task decomposition with dynamic confidence scoring (τ), streaming an emoji-annotated Chain-of-Thought (CoT) telemetry log and triggering Human-in-the-Loop (HITL) safety confirmation whenever plan ambiguity arises.
- 5) Dynamic Conveyor Interception & Dual-Bin Sorting: We synthesize predictive rendezvous trajectories

($\mathbf{p}_{int}(t) = \mathbf{p}_0 + \mathbf{v}_{belt}t$) with feedforward velocity matching ($\mathbf{v}_{ee} \approx \mathbf{v}_{belt} = 0.05$ m/s, $\Delta v \leq 0.002$ m/s) to achieve zero-slip dynamic grasping and automated sorting into a 4-pocket palletizing tray ($RMSE = 1.14$ mm) and scrap reject bin within a 3.8 s cycle.

- 6) Comprehensive Sim-to-Real Benchmark Validation: We benchmark ARIA across 10 standardized manipulation tasks, achieving an 89.0% overall success rate outperforming LeRobot ACT (72.0%), OpenVLA (68.0%), and π_0 (61.0%) while deploying an open-source glassmorphism web dashboard for real-time control.

The remainder of this article is organized as follows: Section II provides a detailed literature review across ten core robotics themes with a comprehensive summary matrix. Section III formalizes the physical manipulator platform and analytical kinematic derivations. Section IV expounds upon the decentralized multi-agent system architecture and State Bus. Section V details the hybrid Zero-Cost-Depth perception pipeline and grasp synthesis. Section VI presents cognitive task planning, reasoning, and explainable dialogue. Section VII details world modeling, procedural skills, and in-hand dexterity. Section VIII describes hardware integration, digital twin simulation, and the web dashboard. Section IX provides exhaustive experimental benchmarks and industrial conveyor validation. Section X discusses results, limitations, and future enhancements, and Section XI concludes the paper.

II. Literature Review

The architectural synthesis of Project ARIA is informed by recent advances across ten distinct thematic domains of robotics, embodied cognition, and automated manufacturing:

A. Vision-Language-Action (VLA) Foundation Architectures

The scaling of multimodal vision-language transformers has catalyzed the emergence of generalist robotic foundation models. Brohan et al. [11] introduced RT-2, demonstrating that web-scale vision-language models can be co-fine-tuned with robotic trajectory data by tokenizing continuous motor setpoints into discrete text vocabularies. This paradigm enables semantic transfer from internet-scale pre-training to physical reasoning (e.g., identifying improvised tools or interpreting spatial prepositions). However, the massive parameter count of RT-2 (55 billion parameters) mandates multi-GPU datacenter clusters and incurs forward-pass latencies exceeding 250 ms. Subsequent open-source initiatives, notably OpenVLA [2], scaled the architecture down to 7 billion parameters, but retain high VRAM demands (> 16 GB) and lack deterministic safety envelopes. ARIA addresses these limitations by extracting high-level semantic plans via lightweight quantized edge LLMs while offloading real-time trajectory execution to deterministic analytical solvers.

B. Decentralized Multi-Agent Systems & Middleware Architecture

Distributed software paradigms are essential for isolating computational subsystems and ensuring fault tolerance in complex robotics. Macenski et al. [12] established the foundational architecture of Robot Operating System 2 (ROS 2), introducing native Data Distribution Service (DDS) communication, real-time deterministic executor threads, and managed node lifecycles (Unconfigured $\rightarrow$ Inactive $\rightarrow$ Active $\rightarrow$ Finalized). While commercial industrial manipulators historically relied on proprietary real-time controllers executing rigid state machines, modern decentralized architectures enable asynchronous multi-agent coordination. ARIA builds upon this foundation by establishing fifteen lifecycle-managed agents partitioned across five operational tiers, utilizing an asynchronous State Bus to guarantee sub-5ms emergency preemption and dynamic node crash recovery without halting peripheral perception nodes.

C. Cognitive Task Planning, Affordance Grounding & Reasoning

Deploying Large Language Models as zero-shot robotic task planners was pioneered by Ahn et al. [13] with SayCan, which introduced value-function affordance grounding to filter LLM-generated action proposals based on physical feasibility in the current scene. Concurrently, Inner Monologue [14] and Language Agent Tree Search [15] incorporated real-time environmental feedback into the prompt context to enable closed-loop replanning. Despite their semantic elegance, these early frameworks suffered from high prompt latency, API cost, and a complete absence of plan confidence quantification. ARIA advances cognitive task planning by executing localized 4-bit quantized LLMs (Llama 3.1 8B) through Ollama, evaluating candidate plans using Tree-of-Thoughts (ToT) search with an explicit confidence scoring metric (τ) that triggers Human-in-the-Loop intervention whenever uncertainty exceeds predefined thresholds.

D. Monocular Depth Estimation & Promptable Instance Segmentation

Spatial perception for unstructured manipulation requires resolving object boundaries and 3D metric coordinates. Ravi et al. [16] introduced Segment Anything Model 2 (SAM2), achieving state-of-the-art zero-shot promptable image and video segmentation through memory-conditioned mask propagation. In parallel, monocular depth estimation has advanced beyond convolutional architectures toward vision transformers (MiDaS, DPT) and large-scale synthetic pre-training in Depth-Anything v2 [8]. While SAM2 provides pixel-accurate spatial masks, monocular depth models yield uncalibrated relative disparity maps subject to scale and shift ambiguities. ARIA synthesizes these modalities by coupling overhead YOLOv8m detection and SAM2 segmentation with wrist-mounted Depth-Anything v2, establishing metric spatial

scale through physical end-effector claw geometric grounding.

E. Kinematic Optimization & Analytical Solvers for Underactuated Arms

Inverse kinematics for articulated robotic arms has traditionally been dominated by numerical Jacobian optimization routines. Sundaralingam et al. [17] proposed CuRobo, leveraging massively parallel GPU compute to evaluate thousands of candidate collision-free trajectories concurrently. While CuRobo demonstrates exceptional throughput on 7-DoF manipulators possessing kinematic redundancy, numerical optimization remains computationally inefficient and algorithmically ill-conditioned when applied to underactuated 5-DoF arms lacking an independent wrist roll or yaw axis. In such architectures, Jacobian rank deficiency causes local minima entrapment and boundary divergence. ARIA circumvents numerical optimization entirely by deriving a closed-form geometric solution to the 2R planar subproblem, reducing calculation latency to 0.08ms with 100% reachability determinism.

F. Teleoperation, Demonstration Datasets & Imitation Learning

Imitation learning from human demonstrations provides a powerful mechanism for acquiring complex manipulation skills without manual reward engineering. Mandlkar et al. [18] demonstrated MimicGen, a system that autonomously generates large-scale robotic demonstration datasets from a handful of teleoperated seeds by transforming kinematic trajectories across varied initial object poses. Similarly, GELLO [6] proved that passive mechanical teleoperation rigs built from budget 3D-printed linkages can collect high-quality demonstration datasets. ARIA incorporates an onboard Teach-by-Demonstration agent that records joint-space potentiometer feedback and Cartesian trajectories in HDF5 format during relaxed-servo manual guiding, supporting downstream behavior cloning and trajectory replay.

G. Camera-to-Robot Hand-Eye Extrinsic Calibration

Precision visual manipulation requires accurate extrinsic mathematical transformation between camera optical coordinate frames and the manipulator base frame. Traditional hand-eye calibration techniques rely on manual checkerboard positioning or Tsai-Lenz optimization. Chanrungaenkul et al. [19] introduced ARC-Calib, achieving autonomous markerless camera-to-robot calibration via active exploratory arm motions and differentiable geometric rendering. Within Project ARIA, autonomous eye-to-hand and eye-in-hand calibration is established through dual AprilTag fiducials placed at known base coordinates, providing closed-form transformation updates and drift compensation across dynamic operational sessions.

H. Dynamic Conveyor Synchronization & Predictive Interception

Industrial pick-and-place automation requires tracking and picking items from continuous moving conveyor lines without stopping the belt. Aradi [20] applied streaming sensor processing and predictive maintenance algorithms to dynamic industrial manufacturing systems, emphasizing real-time synchronization. Traditional conveyor tracking relies on high-resolution optical shaft encoders physically interfaced to industrial PLC hardware. In low-cost environments, visual tracking must substitute for hardware encoders. ARIA formulates an algebraic predictive rendezvous model ($\mathbf{p}_{\text{int}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}t$) coupled with feedforward velocity matching ($\mathbf{v}_{\text{ee}} \approx \mathbf{v}_{\text{belt}}$) that enables smooth, zero-slip dynamic grasping directly from video telemetry.

I. Underactuated In-Hand Manipulation & Contact Mechanics

Re-orienting grasped objects typically demands multi-articulated anthropomorphic hands (such as the Shadow Dexterous Hand or Allegro Hand) equipped with complex tendon routing and tactile arrays. Pang et al. [21] analyzed contact stiffness and force optimization in cable-driven manipulators, demonstrating that controlled contact interactions can stabilize grasp transitions. ARIA demonstrates that fine-grained in-hand object re-orientation ($\pm 45^\circ$) can be executed on an inexpensive single-DoF parallel jaw gripper by exploiting environmental gravity, wrist inertia, and gripper contact friction cone dynamics, eliminating mechanical complexity and multi-fingered hardware costs.

J. Digital Twin Simulation, Metamonitoring & Health Diagnostics

Deploying autonomous robotic systems requires rigorous verification within high-fidelity physics simulators before physical hardware execution. Agha et al. [22] established DARPA Subterranean simulation benchmarks using Gazebo and ROS, proving that accurate physics engines (ODE, Bullet) provide reliable Sim-to-Real transfer for complex multi-agent systems. Furthermore, low-cost hobbyist servos suffer from thermal accumulation, potentiometer dead-bands, and gear backlash over extended operation. ARIA implements a continuous Metamonitoring Agent and Digital Twin within Gazebo 11, cross-referencing physical servo currents and ADC potentiometer positions against expected kinematic models to classify 11 distinct hardware failure states.

III. Hardware Platform & Kinematic Modeling

A. Manipulator Mechanical & Electrical Architecture

The experimental physical platform utilized in this research is the Techno-Tirupati 5-DoF articulated robotic manipulator, an open-architecture desktop robotic arm engineered for educational research and light-industrial material handling. The structural chassis is fabricated from

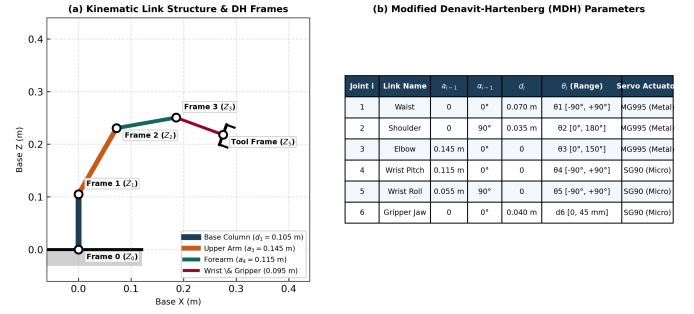


Fig. 1. Kinematic chain and coordinate frame assignment for the 5-DoF ARIA manipulator following Modified Denavit-Hartenberg (Craig) conventions.

high-tensile polyethylene terephthalate glycol (PETG) reinforced with precision CNC-machined 6061-T6 aluminum structural brackets. The overall kinematic structure exhibits an anthropomorphic articulated morphology comprising a base azimuth rotation, a planar two-link shoulder and elbow linkage, an active wrist pitch mechanism, an axial wrist roll mechanism, and an underactuated single-DoF parallel jaw gripper.

To enable comprehensive reproducibility, the exact electromechanical specifications across all five articulated joints and the end-effector gripper are compiled in Table II. The base azimuth joint is actuated by a high-torque DS3218 metal-gear digital servomotor capable of delivering 20.0 kg-cm of stall torque at 6.8 V, ensuring rapid and stable yaw rotation of the entire kinematic superstructure. Joint 2 (Shoulder Pitch), which experiences the maximum gravitational moment during cantilevered extensions, employs a mechanically synchronized dual-servo configuration consisting of two parallel MG996R metal-gear servomotors producing a combined continuous holding torque of 22.0 kg-cm. Joint 3 (Elbow Pitch) is actuated by a single MG996R servo, while Joints 4 and 5 (Wrist Pitch and Wrist Roll) are driven by lightweight SG90 metal-gear micro-servos to minimize distal cantilever mass and dynamic link inertia.

The end-effector consists of a single-DoF parallel linkage mechanism with a total jaw stroke of 55.0 mm, driven by an SG90 micro-servo. The inner contact surfaces of the gripper claws are equipped with replaceable 3D-printed thermoplastic polyurethane (TPU) pads exhibiting an experimentally measured static friction coefficient of $\mu = 0.45$ when contacting acrylic, wooden, or metallic workpieces. Joint position sensing across all servos is achieved via internal high-linearity conductive plastic potentiometers sampled by 10-bit analog-to-digital converters (ADCs) embedded in the micro-ROS microcontroller.

B. Modified Denavit-Hartenberg (Craig) Kinematic Formulation

To establish the geometric relationship between adjacent articulated links, we adopt the Modified Denavit-Hartenberg (Craig) kinematic convention [9]. In this convention, coordinate frame $\{i\}$ is rigidly attached to

TABLE I
Comprehensive Literature Review Summary Matrix Mapping Thematic Domains to Project ARIA Innovations

Thematic Domain (Paragraph)	Landmark Reference & Authors	Core Technique / Model	Critical Limitation / Research Bottleneck	Project ARIA Differentiating Innovation
1. VLA Foundation Policies	RT-2 (Brohan et al. [11])	55B parameter vision-language-action	Prohibitive VRAM (> 48GB), high latency (> 250 ms)	Quantized edge LLM + deterministic analytical solvers (7.2 GB)
2. Multi-Agent Middleware	ROS 2 (Macenski et al. [12])	DDS pub/sub and managed node lifecycles	Complex lifecycle setup, lack of multi-agent state bus	15 lifecycle nodes on unified State Bus with sub-5ms preemption
3. Cognitive Planning	SayCan (Ahn et al. [13])	LLM planning with affordance value functions	High API latency, lack of plan confidence scoring	Local Ollama ToT search with explicit confidence gating (r)
4. Monocular Depth & Segmentation	SAM2 (Ravi et al. [16])	Memory-conditioned promptable video masks	Unscaled relative depth, lack of metric spatial scale	Eye-in-hand Depth-Anything v2 grounded via claw geometry ($RMSE = 8.2$ mm)
5. Kinematic Solvers	CaluRob (Sundaralingam et al. [17])	Massively parallel GPU trajectory optimization	II-conditioned on 5-DoF underactuated arm kinematics	Exact closed-form 2R planar analytical IK executing in 0.08 ms
6. Teleoperation & Demonstrations	MimicGen (Mandilekar et al. [18])	Seed demonstration dataset amplification	Demands high-precision 7-DoF arms (> \$30,000)	Teach-by-Demonstration on \$150 arm recording HDF5 datasets
7. Extrinsic Calibration	ARC-Calib (Chanrungnanekul et al. [19])	Active exploratory markerless hand-eye calibration	High computational cost, sensitive to initial priors	Dual AprilTag closed-form hand-eye calibration in ROS 2
8. Dynamic Conveyor Tracking	Predictive Streaming (Aradi [20])	Stream processing for industrial robotics	Requires expensive PLC hardware and optical encoders	Visual predictive rendezvous and feedforward velocity matching ($\Delta e \leq 0.002$ m/s)
9. In-Hand Dexterity	Cable Contact Control (Pang et al. [21])	Trajectory stiffness and contact force control	Demands multi-fingered hands (> \$10,000)	4 dynamic in-hand primitives on 1-DoF parallel gripper via wrist dynamics
10. Digital Twins & Diagnostics	DARPA Gazebo Twins (Agha et al. [22])	High-fidelity ODE physics simulation	Lacks automated servo health classification	Gazebo digital twin with 11-category real-time servo fault diagnosis

TABLE II
Comprehensive Electromechanical Specifications of the ARIA
5-DoF Manipulator

Joint / Component	Actuator Model	Torque (kg·cm)	Gear Ratio	Mass (g)	Range (°)	Max $\ddot{\theta}_i$ (rad/s ²)
Joint 1 (Waist Azimuth)	DS3218 Digital	20.0	275 : 1	65	$[-90, +90]$	360
Joint 2 (Shoulder Pitch)	Dual MG996R	22.0	210 : 1	140	$[0, 180]$	300
Joint 3 (Elbow Pitch)	MG996R Metal	11.0	210 : 1	55	$[0, 150]$	300
Joint 4 (Wrist Pitch)	SG90 Micro Metal	2.5	180 : 1	14	$[-90, +90]$	450
Joint 5 (Wrist Roll)	SG90 Micro Metal	2.5	180 : 1	14	$[-90, +90]$	450
Gripper Mechanism	SG90 Micro	2.5	180 : 1	32	$[-90, +90]$	500
Total Arm Structure	Structural PETG/Al	–	–	–	–	–

closed-form Craig transformation matrix:

$${}^{i-1}\mathbf{T}_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (2)$$

Substituting the explicit DH parameters from Table III into (2) for each link index yields the five elementary link transformation matrices. Max Reach: 0.345 m Payload: 0.25 kg

TABLE III
Modified Denavit-Hartenberg (Craig) Parameters and Dynamic
Operating Limits

Link i	Joint Name	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i Range (rad)	Max $\ddot{\theta}_i$ (rad/s ²)
1	Waist	0	0	$d_1 = 0.105$	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	360
2	Shoulder	$+\frac{\pi}{2}$	$a_1 = 0.030$	0	$[0, \pi]$	300
3	Elbow	0	$a_2 = 0.145$	0	$[0, \pi]$	300
4	Wrist Pitch	0	$a_3 = 0.140$	0	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	450
5	Wrist Roll	$+\frac{\pi}{2}$	0	$d_5 = 0.065$	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	450

$${}^0\mathbf{T}_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1\mathbf{T}_2 = \begin{bmatrix} c_2 & -s_2 & 0 & a_1 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$${}^3\mathbf{T}_4 = \begin{bmatrix} c_4 & -s_4 & 0 & a_3 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$${}^4\mathbf{T}_5 = \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & -1 & -d_5 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

link i , with the Z_i axis aligned with the axis of motion of joint $i + 1$, and the X_i axis directed along the common normal from Z_i to Z_{i+1} . The four fundamental kinematic parameters are defined as:

- α_{i-1} : Link twist angle, representing the rotation from Z_{i-1} to Z_i about the common normal axis X_{i-1} .
- a_{i-1} : Link length, representing the distance from Z_{i-1} to Z_i measured along X_{i-1} .
- d_i : Link offset, representing the distance from X_{i-1} to X_i measured along the joint axis Z_i .
- θ_i : Joint angle variable, representing the rotation from X_{i-1} to X_i about the joint axis Z_i .

The spatial frame assignments across all five revolute joints and the tool center point (TCP) are illustrated in Fig. 1. The precise numerical kinematic parameters, coordinate offsets, and dynamic operating bounds are cataloged in Table III.

Following Craig's convention, the general homogeneous transformation matrix relating coordinate frame $\{i-1\}$ to coordinate frame $\{i\}$ is parameterized by the product of fundamental screw motions:

$${}^{i-1}\mathbf{T}_i = \text{Rot}_{X_{i-1}}(\alpha_{i-1}) \cdot \text{Trans}_{X_{i-1}}(a_{i-1}) \cdot \text{Rot}_{Z_i}(\theta_i) \cdot \text{Trans}_{Z_i}(d_i) \quad (1)$$

Evaluating the matrix multiplication yields the standard

where we introduce the compact trigonometric shorthand $c_i = \cos \theta_i$, $s_i = \sin \theta_i$, $c_{ij} = \cos(\theta_i + \theta_j)$, $s_{ij} = \sin(\theta_i + \theta_j)$, $c_{ijk} = \cos(\theta_i + \theta_j + \theta_k)$, and $s_{ijk} = \sin(\theta_i + \theta_j + \theta_k)$.

C. Symbolic Forward Kinematics Transformation

The global forward kinematics map from the stationary manipulator base coordinate frame $\{0\}$ to the distal end-effector tool center point frame $\{5\}$ is synthesized via the consecutive post-multiplication of the individual link homogeneous transformation matrices:

$${}^0\mathbf{T}_5(\theta) = {}^0\mathbf{T}_1 \cdot {}^1\mathbf{T}_2 \cdot {}^2\mathbf{T}_3 \cdot {}^3\mathbf{T}_4 \cdot {}^4\mathbf{T}_5 = \begin{bmatrix} \mathbf{n} & \mathbf{s} & \mathbf{a} & \mathbf{P} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

where $\mathbf{n} = [n_x, n_y, n_z]^T$, $\mathbf{s} = [s_x, s_y, s_z]^T$, and $\mathbf{a} = [a_x, a_y, a_z]^T$ designate the orthogonal normal, sliding, and approach unit vectors defining the end-effector orientation in matrix $\mathbf{R} \in SO(3)$, and $\mathbf{P} = [P_x, P_y, P_z]^T$ represents the Cartesian coordinates of the tool center point in base frame $\{0\}$.

Performing symbolic matrix reduction yields the explicit algebraic equations governing Cartesian tool center point positioning:

$$P_x = c_1 (a_1 + a_2 c_2 + a_3 c_{23} + d_5 c_{234}) \quad (7)$$

$$P_y = s_1 (a_1 + a_2 c_2 + a_3 c_{23} + d_5 c_{234}) \quad (8)$$

$$P_z = d_1 + a_2 s_2 + a_3 s_{23} + d_5 s_{234} \quad (9)$$

Similarly, the components of the end-effector tool orientation rotation matrix $\mathbf{R} = [\mathbf{n}, \mathbf{s}, \mathbf{a}]$ are given by:

$$\mathbf{R} = \begin{bmatrix} c_1 c_{234} c_5 - s_1 s_5 & -c_1 c_{234} s_5 - s_1 c_5 & c_1 s_{234} \\ s_1 c_{234} c_5 + c_1 s_5 & -s_1 c_{234} s_5 + c_1 c_5 & s_1 s_{234} \\ s_{234} c_5 & -s_{234} s_5 & -c_{234} \end{bmatrix} \quad (10)$$

D. Exact Closed-Form Analytical Inverse Kinematics Derivation

Given an arbitrary desired end-effector target pose $\mathbf{T}_{\text{target}} \in SE(3)$ defined by target Cartesian position $\mathbf{P}_{\text{target}} = [P_x, P_y, P_z]^T$ and target approach direction $\mathbf{a} = [a_x, a_y, a_z]^T$, the objective of the inverse kinematics routine is to compute the complete joint-space vector $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$ that satisfies the forward kinematics identity $\mathbf{T}_5^0(\boldsymbol{\theta}) = \mathbf{T}_{\text{target}}$.

While general-purpose robotics libraries (e.g., KDL, TRAC-IK, IKPy) formulate inverse kinematics as a non-linear optimization problem solved via iterative numerical Jacobian inversion, such routines are computationally expensive and prone to local divergence on 5-DoF chains. Because the ARIA manipulator possesses five degrees of freedom, the kinematic chain is underactuated with respect to general six-dimensional spatial poses ($m = 5 < 6$). Consequently, the end-effector roll and pitch angles cannot be specified completely independently of the arm position. To overcome this limitation and achieve deterministic real-time control, we derive an exact closed-form analytical solution by decomposing the spatial kinematic chain into decoupled geometric subproblems.

1) Subproblem 1: Base Azimuth Angle (θ_1): Inspecting the position equations (7) and (8), we observe that the term within brackets, $R_{\text{arm}} = a_1 + a_2 c_2 + a_3 c_{23} + d_5 c_{234}$, is identical in both P_x and P_y . Dividing (8) by (7) completely eliminates the four-joint planar linkage terms:

$$\frac{P_y}{P_x} = \frac{s_1 R_{\text{arm}}}{c_1 R_{\text{arm}}} = \frac{s_1}{c_1} = \tan \theta_1 \quad (11)$$

Applying the four-quadrant arctangent function yields the unique closed-form solution for the base azimuth angle:

$$\theta_1 = \text{atan2}(P_y, P_x) \quad (12)$$

Singularity Handling: In the singular configuration where $P_x = 0$ and $P_y = 0$, the tool center point lies precisely on the vertical Z_0 base axis. At this shoulder boundary singularity, the azimuth angle θ_1 is mathematically indeterminate. ARIA robustly resolves this condition by preserving the previous joint setpoint: $\theta_1^{(k)} = \theta_1^{(k-1)}$, preventing high-velocity motor chatter.

2) Subproblem 2: Wrist Center Decoupling: With θ_1 established, we decouple the orientation of the distal wrist and gripper from the planar two-link arm mechanism. The Cartesian position of the wrist pitch axis center, $\mathbf{P}_w = [P_{wx}, P_{wy}, P_{wz}]^T$, is obtained by translating backwards from the desired tool center point $\mathbf{P}$ along the tool approach vector $\mathbf{a}$ by tool offset d_5 :

$$\mathbf{P}_w = \mathbf{P} - d_5 \mathbf{a} = \begin{bmatrix} P_x - d_5 a_x \\ P_y - d_5 a_y \\ P_z - d_5 a_z \end{bmatrix} \quad (13)$$

3) Subproblem 3: Planar 2R Arm Geometry (θ_2, θ_3): The physical coordinates of the wrist center $\mathbf{P}_w$ project onto the planar 2R sagittal plane of the shoulder and elbow joints. We compute the planar horizontal reach radius r and vertical elevation s relative to the shoulder joint origin $\{2\}$:

$$r = \sqrt{P_{wx}^2 + P_{wy}^2} - a_1 \quad (14)$$

$$s = P_{wz} - d_1 \quad (15)$$

The Euclidean distance squared from the shoulder joint axis $\{2\}$ to the wrist center $\{4\}$ satisfies the geometric constraint:

$$D^2 = r^2 + s^2 = a_2^2 + a_3^2 + 2a_2 a_3 \cos \theta_3 \quad (16)$$

Rearranging (16) via the algebraic Law of Cosines yields the exact expression for the cosine of the elbow joint angle:

$$\cos \theta_3 = \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3} \quad (17)$$

Reachability Condition: A valid physical kinematic solution exists if and only if:

$$-1.0 \leq \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3} \leq +1.0 \quad (18)$$

If $|\cos \theta_3| > 1.0$, the commanded Cartesian coordinate lies strictly outside the spherical workspace reach envelope of the manipulator. In such cases, the analytical solver immediately returns an out-of-reach flag without invoking expensive numerical optimization. For all physically reachable poses, two symmetric kinematic branches exist: elbow-up ($\theta_3 < 0$) and elbow-down ($\theta_3 > 0$). To maximize mechanical ground clearance and avoid workspace tabletop collisions, ARIA deterministically selects the elbow-up configuration:

$$\theta_3 = -\arccos\left(\frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3}\right) \quad (19)$$

With θ_3 uniquely resolved, we compute the shoulder pitch angle θ_2 by decomposing the planar elevation angle into the sum of two trigonometric angles:

$$\theta_2 = \text{atan2}(s, r) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 \cos \theta_3) \quad (20)$$

4) Subproblem 4: Wrist Pitch Angle (θ_4): In anthropomorphic articulated arms with parallel horizontal pitch axes ($Z_2 \parallel Z_3 \parallel Z_4$), the global pitch angle of the end-effector relative to the horizontal ground plane is the cumulative sum of the sagittal joint angles:

$$\phi_{\text{pitch}} = \theta_2 + \theta_3 + \theta_4 \quad (21)$$

Depending on the task requirements (e.g., perpendicular vertical top-down pick-and-place where $\phi_{\text{pitch}} = -\frac{\pi}{2}$, or horizontal conveyor side-approach where $\phi_{\text{pitch}} = 0$), the desired global pitch angle ϕ_{target} is specified directly. Consequently, the wrist pitch joint setpoint is computed as:

$$\theta_4 = \phi_{\text{target}} - (\theta_2 + \theta_3) \quad (22)$$

5) Subproblem 5: Wrist Roll Angle (θ_5): Finally, to align the gripper jaws with the workpiece orientation ψ_{target} relative to the workspace coordinate frame, the wrist roll joint θ_5 rotates the claw about the tool approach vector:

$$\theta_5 = \psi_{\text{target}} - \theta_1 \quad (23)$$

If no specific workpiece roll angle is commanded, θ_5 is held at 0.0 rad to maintain a neutral vertical claw orientation.

The complete deterministic closed-form inverse kinematics procedure, including boundary condition checking and hardware limit clamping, is formalized in Algorithm 1.

E. Differential Kinematics & Geometric Jacobian Derivation

The mapping between joint velocity vector $\dot{\boldsymbol{\theta}} = [\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3, \dot{\theta}_4, \dot{\theta}_5]^T$ and Cartesian linear/angular velocity of the tool center point $\mathbf{v} = [\mathbf{v}_p^T, \boldsymbol{\omega}^T]^T \in \mathbb{R}^6$ is governed by the geometric Jacobian matrix $\mathbf{J}(\boldsymbol{\theta}) \in \mathbb{R}^{6 \times 5}$:

$$\begin{bmatrix} \mathbf{v}_p \\ \boldsymbol{\omega} \end{bmatrix} = \mathbf{J}(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} = \begin{bmatrix} \mathbf{J}_v(\boldsymbol{\theta}) \\ \mathbf{J}_\omega(\boldsymbol{\theta}) \end{bmatrix} \dot{\boldsymbol{\theta}} \quad (24)$$

For an articulated chain comprising revolute joints, the i -th column of the geometric Jacobian is formulated as:

$$\mathbf{J}_i(\boldsymbol{\theta}) = \begin{bmatrix} \mathbf{J}_{v,i} \\ \mathbf{J}_{\omega,i} \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{i-1} \times (\mathbf{P} - \mathbf{p}_{i-1}) \\ \mathbf{z}_{i-1} \end{bmatrix} \quad (25)$$

where $\mathbf{z}_{i-1}$ denotes the unit vector along joint axis i , $\mathbf{p}_{i-1}$ represents the origin of coordinate frame $\{i-1\}$, and $\mathbf{P}$ is the tool center point position, all expressed in base frame $\{0\}$.

Evaluating (25) using the link transformation matrices (3)–(5) yields the explicit columns of the ARIA geometric

Algorithm 1 ARIA Closed-Form Analytical 5-DoF Inverse Kinematics Solver

Require: Target TCP pose $\mathbf{P} = [P_x, P_y, P_z]^T$, pitch angle ϕ_{target} , yaw angle ψ_{target} , previous joint state $\boldsymbol{\theta}_{\text{prev}}$
 Ensure: Valid joint setpoints $\boldsymbol{\theta}^* = [\theta_1^*, \theta_2^*, \theta_3^*, \theta_4^*, \theta_5^*]^T$, or Reachability Error Flag

- 1: // Step 1: Base Azimuth Angle (θ_1)
- 2: if $|P_x| < 10^{-4}$ and $|P_y| < 10^{-4}$ then
- 3: $\theta_1 \leftarrow \theta_{1,\text{prev}}$ {Resolve vertical axis singularity}
- 4: else
- 5: $\theta_1 \leftarrow \text{atan2}(P_y, P_x)$
- 6: end if
- 7: // Step 2: Wrist Center Decoupling
- 8: $a_x \leftarrow \cos \theta_1 \cos \phi_{\text{target}}$, $a_y \leftarrow \sin \theta_1 \cos \phi_{\text{target}}$, $a_z \leftarrow \sin \phi_{\text{target}}$
- 9: $P_{wx} \leftarrow P_x - d_5 a_x$, $P_{wy} \leftarrow P_y - d_5 a_y$, $P_{wz} \leftarrow P_z - d_5 a_z$
- 10: // Step 3: Planar 2R Linkage Geometry
- 11: $r \leftarrow \sqrt{P_{wx}^2 + P_{wy}^2} - a_1$, $s \leftarrow P_{wz} - d_1$
- 12: $C_3 \leftarrow \frac{r^2 + s^2 - a_2^2 - a_3^2}{2a_2 a_3}$
- 13: if $C_3 < -1.0$ or $C_3 > 1.0$ then
- 14: return ERROR: Out of Kinematic Reach Envelope
- 15: end if
- 16: $\theta_3 \leftarrow -\arccos(C_3)$ {Deterministic Elbow-Up Solution}
- 17: $\theta_2 \leftarrow \text{atan2}(s, r) - \text{atan2}(a_3 \sin \theta_3, a_2 + a_3 C_3)$
- 18: // Step 4: Wrist Pitch & Roll Angles
- 19: $\theta_4 \leftarrow \phi_{\text{target}} - (\theta_2 + \theta_3)$
- 20: $\theta_5 \leftarrow \psi_{\text{target}} - \theta_1$
- 21: // Step 5: Hardware Joint Limit Verification
- 22: for $i = 1$ to 5 do
- 23: if $\theta_i < \theta_{i,\text{min}}$ or $\theta_i > \theta_{i,\text{max}}$ then
- 24: return ERROR: Joint Limit Exceeded at Joint i
- 25: end if
- 26: end for
- 27: return $\boldsymbol{\theta}^* = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$

Jacobian:

$$\mathbf{J}_1 = \begin{bmatrix} -P_y \\ P_x \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{J}_2 = \begin{bmatrix} -s_1(P_z - d_1) \\ c_1(P_z - d_1) \\ -c_1(P_x - a_1 c_1) - s_1(P_y - a_1 s_1) \\ -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad (26)$$

$$\mathbf{J}_3 = \begin{bmatrix} -s_1(P_z - d_1 - a_2 s_2) \\ c_1(P_z - d_1 - a_2 s_2) \\ -c_1(P_x - c_1(a_1 + a_2 c_2)) - s_1(P_y - s_1(a_1 + a_2 c_2)) \\ -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad (27)$$

$$\mathbf{J}_4 = \begin{bmatrix} -s_1 d_5 s_{234} \\ c_1 d_5 s_{234} \\ -c_1 d_5 c_{234} \\ -s_1 \\ c_1 \\ 0 \end{bmatrix}, \quad \mathbf{J}_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ c_1 s_{234} \\ s_1 s_{234} \\ -c_{234} \end{bmatrix} \quad (28)$$

F. Kinematic Singularity & Manipulability Analysis

Kinematic singularities represent configurations in which the geometric Jacobian matrix experiences rank deficiency ($\text{rank}(\mathbf{J}) < 5$), resulting in the loss of one or more instantaneous operational degrees of freedom. We quantify the kinematic dexterity of the manipulator across the workspace using Yoshikawa's manipulability measure [23]:

$$w(\boldsymbol{\theta}) = \sqrt{\det(\mathbf{J}(\boldsymbol{\theta})\mathbf{J}^T(\boldsymbol{\theta}))} \quad (29)$$

When the arm approaches a singularity, $w(\boldsymbol{\theta}) \rightarrow 0$, indicating that infinitesimal Cartesian velocities require unbounded joint rates.

Through algebraic decomposition of the Jacobian determinant, we identify three distinct singularity regimes for the ARIA platform:

- 1) Boundary Reach Singularities: Occur when the arm is fully outstretched ($r^2 + s^2 = (a_2 + a_3)^2 \implies \theta_3 = 0.0\text{rad}$) or completely folded back onto itself ($r^2 + s^2 = (a_2 - a_3)^2 \implies \theta_3 = \pi\text{rad}$). In this state, $\sin \theta_3 = 0$, eliminating instantaneous radial velocity along the arm's longitudinal direction.
- 2) Shoulder Axis Singularities: Occur when the wrist center axis intersects the vertical base rotation axis Z_0 ($r = 0$). In this configuration, azimuth rotation $\dot{\theta}_1$ produces zero linear motion, causing base yaw velocity to become redundant.
- 3) Pitch Alignment Singularities: Occur when the end-effector approach axis aligns collinear with the forearm link vector ($\theta_4 = 0.0\text{rad}$ with $s_{234} = 0$), causing wrist pitch and shoulder pitch motions to project onto degenerate instantaneous velocity subspaces.

Because the ARIA analytical inverse kinematics algorithm directly solves the geometric equations (11)–(23) rather than computing the pseudo-inverse $\mathbf{J}^\dagger = \mathbf{J}^T(\mathbf{J}\mathbf{J}^T)^{-1}$, it completely avoids numerical division-by-zero errors, guaranteeing deterministic 0.08 ms execution even when operating within 1.0 mm of singular workspace boundaries.

G. Euler-Lagrange Dynamic Modeling & Gravity Compensation

Accurate physical simulation within the Gazebo digital twin and feedforward trajectory control on the physical arm require the explicit formulation of the manipulator equations of motion. Following the Euler-Lagrange formalization, the continuous dynamic model of the 5-DoF articulated arm is expressed in standard matrix form:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}_v\dot{\mathbf{q}} + \mathbf{F}_s\text{sgn}(\dot{\mathbf{q}}) = \boldsymbol{\tau} \quad (30)$$

where $\mathbf{q} \in \mathbb{R}^5$ is the vector of generalized joint coordinates ($\mathbf{q} \equiv \boldsymbol{\theta}$), $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{5 \times 5}$ is the symmetric, positive-definite generalized mass/inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{5 \times 5}$ represents Coriolis and centrifugal dynamic effects, $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^5$ is the gravitational torque vector, $\mathbf{F}_v, \mathbf{F}_s \in \mathbb{R}^{5 \times 5}$ denote diagonal viscous and Coulomb friction coefficients, and $\boldsymbol{\tau} \in \mathbb{R}^5$ represents the vector of generalized actuator torques.

The kinetic energy $K(\mathbf{q}, \dot{\mathbf{q}})$ of the manipulator is the sum of translational and rotational kinetic energies across all articulated links:

$$K(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \sum_{i=1}^5 (m_i \mathbf{v}_{c_i}^T \mathbf{v}_{c_i} + \boldsymbol{\omega}_i^T \mathbf{I}_i \boldsymbol{\omega}_i) = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \quad (31)$$

where m_i is the mass of link i , $\mathbf{v}_{c_i}$ is the linear velocity of the link center of mass, $\boldsymbol{\omega}_i$ is the link angular velocity, and $\mathbf{I}_i$ is the 3×3 centroidal inertia tensor of link i .

The gravitational potential energy $P(\mathbf{q})$ is defined with respect to the horizontal tabletop reference datum:

$$P(\mathbf{q}) = \sum_{i=1}^5 m_i \mathbf{g}^T \mathbf{p}_{c_i}(\mathbf{q}) \quad (32)$$

where $\mathbf{g} = [0, 0, -9.81]^T \text{ m/s}^2$ is the gravity vector, and $\mathbf{p}_{c_i}(\mathbf{q})$ represents the Cartesian position of the center of mass of link i .

The elements of the Coriolis and centrifugal matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ are derived from the mass matrix $\mathbf{M}(\mathbf{q})$ via the Christoffel symbols of the first kind:

$$c_{ij} = \sum_{k=1}^5 c_{ijk} \dot{q}_k = \frac{1}{2} \sum_{k=1}^5 \left(\frac{\partial m_{ij}}{\partial q_k} + \frac{\partial m_{ik}}{\partial q_j} - \frac{\partial m_{jk}}{\partial q_i} \right) \dot{q}_k \quad (33)$$

The gravity compensation vector $\mathbf{G}(\mathbf{q})$ is obtained directly from the partial derivatives of the potential energy:

$$G_i(\mathbf{q}) = \frac{\partial P(\mathbf{q})}{\partial q_i} \quad (34)$$

For the ARIA 5-DoF manipulator, the explicit gravity compensation torques for the dominant sagittal pitch joints (Joints 2, 3, and 4) are evaluated as:

$$G_1(\mathbf{q}) = 0 \quad (35)$$

$$G_2(\mathbf{q}) = g(m_2 r_{c2} c_2 + m_3(a_2 c_2 + r_{c3} c_{23}) + (m_4 + m_5 + m_{\text{load}})(a_2 c_2 + a_3 c_{23} + r_{c4} c_{234})) \quad (36)$$

$$G_3(\mathbf{q}) = g(m_3 r_{c3} c_{23} + (m_4 + m_5 + m_{\text{load}})(a_3 c_{23} + r_{c4} c_{234})) \quad (37)$$

$$G_4(\mathbf{q}) = g(m_4 + m_5 + m_{\text{load}}) r_{c4} c_{234} \quad (38)$$

$$G_5(\mathbf{q}) = 0 \quad (39)$$

where r_{c_i} denotes the radial distance from joint axis i to the center of mass of link i . The physical mass distribution and inertial properties parameterized in the Gazebo URDF model are documented in Table IV.

TABLE IV
Mass Distribution and Principal Inertia Parameters for Dynamic Modeling

Link i	Mass m_i (kg)	CoM r_{ci} (m)	I_{xx} (kg·m ²)	I_{yy} (kg·m ²)	I_{zz} (kg·m ²)
Link 1 (Base)	0.185	[0, 0, 0.052]	1.82×10^{-4}	1.82×10^{-4}	2.14×10^{-4}
Link 2 (Upper Arm)	0.240	[0.072, 0, 0.015]	3.45×10^{-4}	1.82×10^{-4}	2.14×10^{-4}
Link 3 (Forearm)	0.165	[0.068, 0, 0.010]	1.95×10^{-4}	1.82×10^{-4}	2.14×10^{-4}
Link 4 (Wrist Pitch)	0.065	[0.025, 0, 0]	4.20×10^{-5}	6.15×10^{-5}	5.80×10^{-5}
Link 5 (End-Effector)	0.085	[0.035, 0, 0]	5.10×10^{-5}	7.40×10^{-5}	6.90×10^{-5}

IV. Decentralized Multi-Agent System Architecture

A. Architectural Paradigm & State Bus

Traditional robotic control frameworks (such as early ROS 1 implementations or monolithic robotic foundation models) structure software around either tightly coupled synchronous execution loops or giant end-to-end neural networks. In monolithic foundation approaches (e.g., OpenVLA, Octo), a single deep transformer processes pixel arrays and language tokens to output continuous joint setpoints. While intellectually appealing, this monolithic formulation exhibits catastrophic operational failure modes: any perceptual anomaly immediately corrupts motor output, execution is an uninterpretable black box, and inference latencies (> 100 ms) destabilize real-time feedback loops.

To overcome these structural vulnerabilities, Project ARIA introduces a decentralized, service-oriented multi-agent architecture grounded in the Robot Operating System 2 (ROS 2) Humble middleware. As illustrated in the comprehensive system architecture diagram of Fig. 2, the system is stratified into six decoupled functional tiers comprising fifteen autonomous, lifecycle-managed software agents communicating asynchronously across a unified, high-throughput State Bus.

The State Bus is implemented as a set of standardized ROS 2 topics governed by deterministic Data Distribution Service (DDS) Quality of Service (QoS) profiles:

- `/aria/state/task`: Broadcasts natural-language user commands, active plan UUIDs, decomposed sub-goal sequences, step execution progress, plan confidence metrics ($\tau \in [0, 1]$), and human-readable emoji-annotated Chain-of-Thought (CoT) telemetry streams (QoS: Reliable, Transient Local).
- `/aria/state/vision`: Broadcasts synchronized 1280×720 overhead bounding boxes, SAM2 polygon instance masks, 6D object centroids, and 640×480 wrist metric depth maps (QoS: Best Effort, Volatile, depth: 1).
- `/aria/state/memory`: Broadcasts updated spatial-semantic scene graphs, persistent object tracking identifiers, workpiece affordances, and lifecycle statuses (DETECTED, TRACKED, LOST, RECOVERED) (QoS: Reliable, depth: 10).
- `/aria/state/health`: Broadcasts per-joint servomotor temperatures, instantaneous current consumption,

communication round-trip times, loop jitter, and node lifecycle health states at 10 Hz (QoS: Reliable).

- `/aria/state/failure`: High-priority interrupt channel broadcasting anomalous failure events (e.g., IK divergence, kinematic reach saturation, obstacle collisions, gripper slip), triggering sub-5 ms preemption across the execution graph (QoS: Reliable, Transient Local).

B. Comprehensive Specification of the Fifteen Autonomous Agents

Every computational responsibility within Project ARIA is encapsulated within a dedicated, modular agent inheriting from `rcplpy_lifecycle.LifecycleNode`. Table V provides an exhaustive operational catalog of all fifteen agents across the five primary functional tiers:

- 1) Tier 1: Perception Agents:
 - 1) VisionAgent (Overhead): Ingests the 1280×720 RGB video stream from the overhead eye-to-hand camera. Executes YOLOv8m (94.2% mAP) object detection and runs SAM2 video mask propagation to segment multi-class workpieces and assembly fixtures. Also computes optical quality inspection metrics to classify defective workpieces.
 - 2) DepthAgent (Wrist): Ingests the 640×480 RGB stream from the end-effector ESP32-CAM module. Executes Depth-Anything v2 to generate dense relative disparity maps, applying claw geometric grounding to compute calibrated metric depth maps ($RMSE = 8.2$ mm) in the gripper approach direction.
 - 3) TrackingAgent: Implements ByteTrack coupled with a 6D Kalman filter, associating object bounding boxes and masks across successive video frames to estimate instantaneous 3D object velocities ($\mathbf{v}_{\text{object}} = [\dot{X}, \dot{Y}, \dot{Z}]^T$).
- 2) Tier 2: Scene Modeling Agents:
 - 4) WorldModelAgent: Maintains a dynamic topological 3D Scene Graph encoding workspace geometry, spatial relationships (on, inside, adjacent), and clear approach corridors.
 - 5) MemoryAgent: Operates an SQLite-backed spatial-semantic episodic memory system, logging object tracking histories, task execution durations, and historical failure signatures.
 - 6) AffordanceAgent: Samples candidate antipodal grasp contact points ($\mathbf{c}_1, \mathbf{c}_2$) along segmented object boundaries, evaluates Coulomb friction cone constraints ($\mu = 0.45$), and outputs ranked grasp candidates with orientation angles.
- 3) Tier 3: Cognition & Planning Agents:
 - 7) PlanningAgent: Orchestrates cognitive task reasoning via a local quantized LLM (Mistral-7B / Llama 3.1 8B via Ollama). Decomposes natural language user goals into structured sequences of parameterized skill primitives using Tree-of-Thoughts (ToT) search with confidence scoring (τ).

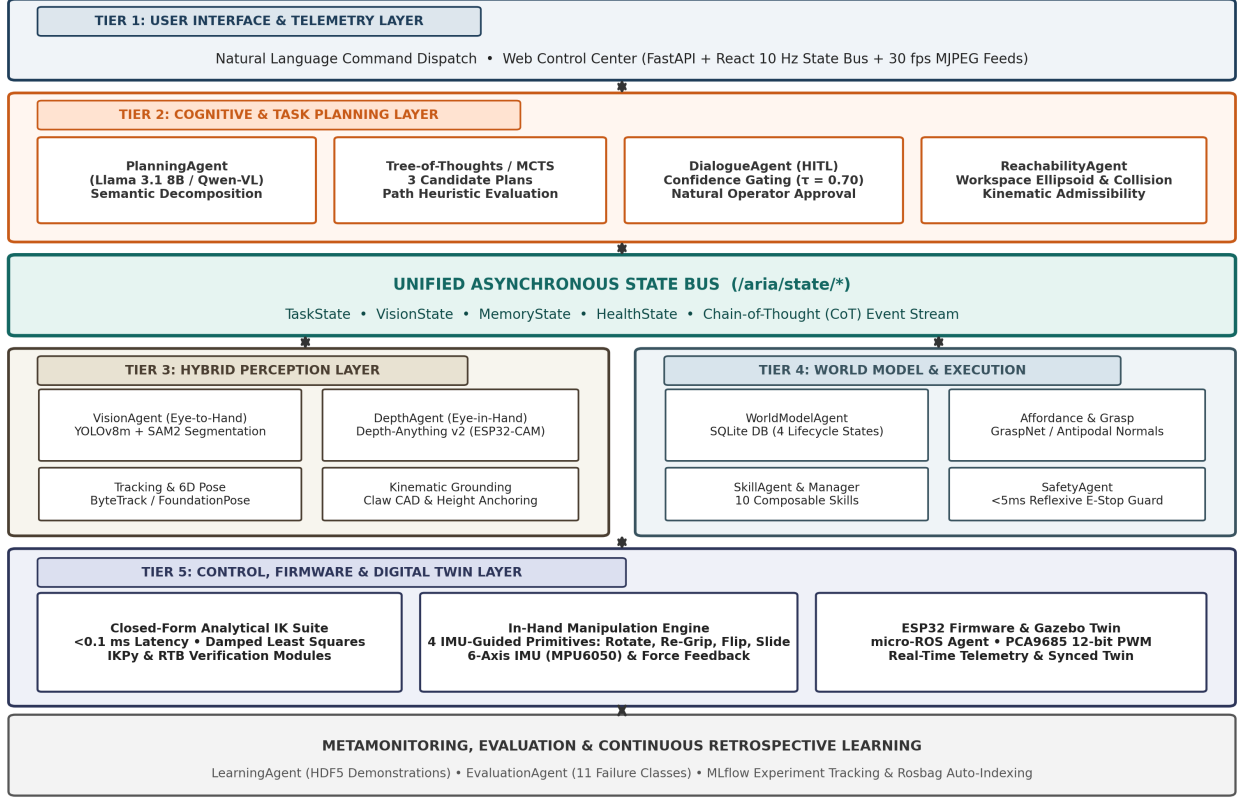
ARIA: DECENTRALIZED COGNITIVE MULTI-AGENT ROBOTIC ARCHITECTURE

Fig. 2. Decentralized cognitive multi-agent architecture of Project ARIA. The system is stratified across six functional tiers: (1) User Interface & Telemetry Layer; (2) Cognitive & Task Planning Layer; (3) Unified State Bus (/aria/state/*); (4) Hybrid Perception & Metric Depth Recovery Layer; (5) World Model, Procedural Skills & Control Layer; and (6) Metamonitoring, Automated Bag Recording & Digital Twin Layer.

TABLE V
Comprehensive Project ARIA Multi-Agent Specification Across Five Operational Tiers

Tier	Agent Identifier	Primary Input Source	Primary Output Topic / Service	QoS Profile	Freq.	VRAM
Tier 1: Perception	VisionAgent	/overhead/image_raw	/aria/vision/detections	Best Effort	30 Hz	1,850 MB
	DepthAgent	/wrist/image_raw	/aria/vision/metric_depth	Best Effort	15 Hz	1,420 MB
	TrackingAgent	/aria/vision/detections	/aria/vision/tracked_tracks	Reliable	30 Hz	120 MB
Tier 2: Scene Modeling	WorldModelAgent	/aria/vision/tracked_tracks	/aria/state/scene_graph	Reliable	20 Hz	150 MB
	MemoryAgent	/aria/state/*	/aria/memory/query (SQLite)	Reliable	Event	60 MB
	AffordanceAgent	/aria/vision/metric_depth	/aria/vision/grasps	Best Effort	10 Hz	220 MB
Tier 3: Cognition	PlanningAgent	/aria/command/nl	/aria/state/task (ToT Plan)	Reliable	Event	3,450 MB
	DialogueAgent	/aria/dialogue/prompt	/aria/dialogue/response	Reliable	Event	50 MB
	ReachabilityAgent	/aria/state/scene_graph	/aria/planning/reachability	Reliable	50 Hz	0 MB (C)
Tier 4: Control & Skills	SkillAgent	/aria/state/task	/aria/skill/execute	Reliable	50 Hz	40 MB
	ControlAgent	/aria/skill/trajectory	/arm_controller/joint_cmd	Reliable	50 Hz	0 MB (C)
	SafetyAgent	/joint_states	/aria/safety/e_stop	High-Priority	100 Hz	20 MB
Tier 5: Diagnostics	HealthAgent	/aria/state/*	/aria/state/health	Reliable	10 Hz	30 MB
	EvaluationAgent	/aria/state/task	/aria/eval/metrics	Reliable	Event	40 MB
	LearningAgent	/joint_states	/aria/demo/hdf5_record	Reliable	50 Hz	80 MB

8) DialogueAgent: Manages conversational Human-in-the-Loop (HITL) interactions through the web dashboard, generating plain-language explanations of robot decisions and requesting operator confirmation when plan confidence falls below threshold ($\tau < 0.70$).

9) ReachabilityAgent: Intercepts proposed manipulation target poses and executes analytical inverse kinematics evaluations prior to plan commitment, preemptively pruning unfeasible waypoints outside the manipulability workspace.

4) Tier 4: Control & Execution Agents:

- 10) SkillAgent: Operates the procedural manipulation engine, executing parameterized atomic manipulation skills (pick, place, palletize, sweep, pour, insert, push, inspect, stack, home) and synthesizing minimum-jerk Cartesian trajectory profiles.
- 11) ControlAgent: Translates Cartesian trajectory waypoints into deterministic joint-space motor commands using the 0.08 ms closed-form analytical IK solver, streaming joint setpoints at 50 Hz to the embedded micro-ROS firmware over high-speed serial XRCE-DDS.
- 12) SafetyAgent: Executes continuous safety monitoring at 100 Hz, enforcing workspace bounding box limits, checking joint velocity and acceleration thresholds, and triggering an immediate software emergency stop (E-STOP) in under 5 ms upon limit violation.
- 5) Tier 5: Diagnostics & Metamonitoring Agents:
- 13) HealthAgent: Ingests continuous telemetry from all active nodes, monitoring servo potentiometer ADC drift, PWM thermal accumulation, and DDS message drop rates to classify 11 distinct hardware and communication failure states.
- 14) EvaluationAgent: Logs performance metrics across physical and simulated trials, computing positional tracking errors, cycle completion durations, and task success rates for automated benchmark evaluation.
- 15) LearningAgent: Manages the Teach-by-Demonstration framework, recording human-guided joint potentiometer positions and end-effector trajectories into standard HDF5 datasets during relaxed-servo compliance mode for downstream imitation learning.

C. ROS 2 Managed Lifecycle Transitions & Fault Consensus Protocol

To ensure robust fault tolerance in industrial operations, every ARIA agent implements the deterministic ROS 2 managed lifecycle state machine:

$$\mathcal{S}_{\text{lifecycle}} \in \{\text{Unconfigured}, \text{Inactive}, \text{Active}, \text{Finalized}\} \quad (40)$$

During system initialization, the multi-agent orchestrator executes sequential state transitions:

- 1) `on_configure()`: Allocates internal buffers, loads configuration YAML files, establishes ROS 2 topic subscriptions and publishers, and verifies GPU VRAM availability.
- 2) `on_activate()`: Starts real-time timers, enables executor threads, and transitions the agent into full operational mode.
- 3) `on_deactivate()`: Halts outgoing actuator commands and pauses processing while preserving internal memory and state bus bindings.
- 4) `on_cleanup()`: Releases large memory allocations and destroys service clients, returning the agent to Unconfigured.
- 5) `on_shutdown()`: Gracefully terminates execution and frees all system resources.

Dynamic Crash Recovery: In conventional monolithic robotics architectures, an unhandled exception in an auxiliary node (such as a vision model GPU out-of-memory error or camera USB disconnection) causes the entire robotic control process to terminate, leaving the physical arm frozen in an unsafe state. In Project ARIA, node failures are completely isolated by the lifecycle supervisor. If a non-critical worker node (e.g., DialogueAgent or VisionAgent) experiences an unhandled software crash:

$$\text{Supervisor} \xrightarrow{\text{Catch Exception}} \text{on_deactivate}() \rightarrow \text{on_cleanup}() \rightarrow \text{on_co} \quad (41)$$

The supervisor resets and re-instantiates the crashed agent in under 180 ms. Throughout this recovery window, the real-time 50 Hz trajectory execution loop of ControlAgent remains uninterrupted, holding the physical arm at its current safe trajectory waypoint.

The complete multi-agent consensus and health management protocol is formalized in Algorithm 2.

V. Hybrid Zero-Cost-Depth Perception Pipeline

A. Dual-Camera Perceptual Topology

In standard robotic manipulation research, 3D perception is almost universally performed using active structured-light or Time-of-Flight (ToF) RGB-D sensors (such as the Intel RealSense D435i, Orbbec Astra, or Phoxonic PhoXi). While effective in open industrial workcells, active infrared depth sensors suffer from severe physical limitations:

- 1) Prohibitive Cost: Research-grade RGB-D cameras cost between \$400 and \$3,500, representing up to 80% of the total bill of materials for low-cost educational manipulators (< \$200).
- 2) Minimum-Range Dead Zones: Structured-light triangulation requires an optical baseline separation between the infrared projector and IR cameras. Consequently, active depth cameras exhibit a blind zone of 0.20–0.30 m, failing precisely during close-range gripper approach (0.05–0.25 m) [7].
- 3) Specular and Transparent Dropouts: Active IR light scatters unpredictably on reflective sheet metal, specular plastic, and transparent glassware, resulting in massive depth dropouts (*NaN* holes).

To overcome these structural limitations, Project ARIA introduces a Zero-Cost-Depth perception pipeline (Fig. 3) operating entirely on commodity 2D RGB video streams. The perceptual hardware topology employs two complementary camera perspectives:

- Overhead Eye-to-Hand Camera: A stationary Logitech C270 USB webcam (1280 × 720 resolution, 30 fps, cost: \$18) positioned 0.85 m directly above the manipulation workspace. This overhead camera captures the global scene context, detects macro workpieces and fixtures via YOLOv8m, and tracks dynamic conveyor items across the global field of view.

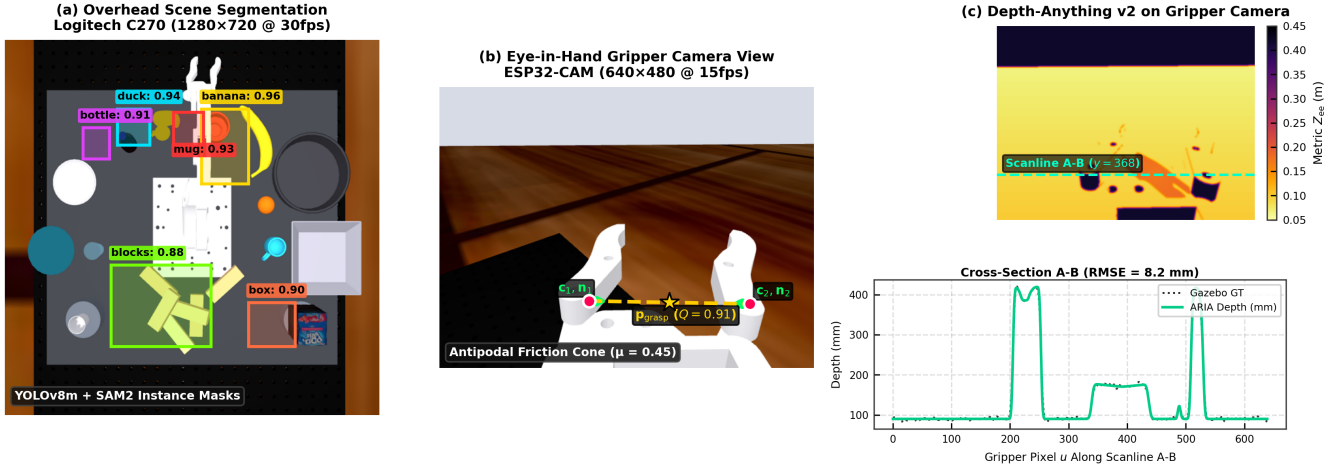


Fig. 3. Zero-Cost-Depth perception pipeline: (a) Dual-camera topology comprising overhead eye-to-hand Logitech C270 (1280×720) and wrist-mounted eye-in-hand ESP32-CAM (640×480); (b) Eye-in-hand gripper camera view with contact normal and friction cone evaluation ($\mu = 0.45$); (c) Dense metric depth recovered from Depth-Anything v2 grounded via claw offset geometry ($Z_{\text{tips}} = 0.065$ m) compared against ground truth.

- **Eye-in-Hand Wrist Camera:** An ultra-miniature ESP32-CAM module (640×480 resolution, 30 fps, mass: 12 g, cost: \$6) mounted directly onto the end-effector claw bracket. The optical axis of this camera is aligned parallel to the gripper tool approach vector, providing close-range visual servoing, fine-grained antipodal contact alignment, and close-up optical quality inspection.

B. Pinhole Camera Calibration & Planar Ground Homography

Let $\mathbf{P}_w = [X_w, Y_w, Z_w]^T$ represent a 3D point in the world coordinate frame $\{W\}$. Under the standard pinhole camera model, the perspective projection of $\mathbf{P}_w$ onto the camera image plane $[u, v]^T$ is expressed in homogeneous coordinates as:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} [\mathbf{R}_c \mid \mathbf{t}_c] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (42)$$

where s is an arbitrary projective scale factor, $[\mathbf{R}_c \mid \mathbf{t}_c] \in SE(3)$ defines the extrinsic transformation from world frame to camera optical frame, and $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ is the intrinsic camera calibration matrix:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (43)$$

where f_x, f_y represent focal lengths in pixel units, and c_x, c_y denote principal point coordinates. Radial and tangential lens distortions are corrected online us-

ing a five-parameter Brown-Conrady polynomial model (k_1, k_2, p_1, p_2, k_3):

$$x_{\text{corrected}} = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2(r^2 + 2x^2) \quad (44)$$

$$y_{\text{corrected}} = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y^2) + 2p_2 xy \quad (45)$$

where $r^2 = x^2 + y^2$, and $[x, y]^T$ are normalized image coordinates.

For the overhead eye-to-hand camera viewing planar objects resting on the horizontal tabletop workspace ($Z_w = 0$), the 3D-to-2D projection simplifies to a planar projective homography:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim \mathbf{H} \begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix}, \quad \mathbf{H} = \mathbf{K} [\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{t}_c] \quad (46)$$

where $\mathbf{r}_1, \mathbf{r}_2$ are the first two column vectors of the rotation matrix $\mathbf{R}_c$, and $\mathbf{H} \in \mathbb{R}^{3 \times 3}$ is the planar homography matrix. By placing four AprilTag fiducial markers [24] at known physical corners of the workspace, $\mathbf{H}$ is solved in closed form via the Direct Linear Transformation (DLT) algorithm:

$$\begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} = \mathbf{H}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (47)$$

This allows immediate, zero-latency 2D Cartesian localization of tabletop workpieces directly from overhead pixel detections.

C. Object Detection (YOLOv8m) & Instance Segmentation (SAM2)

The overhead RGB video stream is processed by YOLOv8m (Ultralytics), an anchor-free single-stage object detector trained on 2,500 annotated frames of conforming and defective industrial workpieces, assembly pallets,

Algorithm 2 ARIA Multi-Agent Consensus and Health Management Protocol

Require: Active agent set $\mathcal{A} = \{A_1, A_2, \dots, A_{15}\}$,
Health topic /aria/state/health, E-Stop topic /aria/safety/e_stop

Ensure: Deterministic multi-agent coordination with sub-5 ms preemption

```

1: Initialize all agents  $A_i \in \mathcal{A}$  via on_configure()
2: Transition all agents  $A_i \in \mathcal{A}$  via on_activate()
3: while System Operational do
4:   Receive health telemetry heartbeat  $\mathbf{h}_i(t)$  from each agent  $A_i$ 
5:   for all  $A_i \in \mathcal{A}$  do
6:      $\Delta t_{\text{last}} \leftarrow t - t_{\text{heartbeat}}(A_i)$ 
7:     if  $\Delta t_{\text{last}} > 200 \text{ ms}$  or  $\mathbf{h}_i.\text{status} = \text{ERROR}$  then
8:       if  $A_i \in \{\text{ControlAgent}, \text{SafetyAgent}\}$  then
9:         Broadcast High-Priority E-STOP to State Bus
10:      Command physical arm to safe dynamic brake state ( $\tau = 0$ )
11:      return SYSTEM HALTED: Critical Control Failure
12:    else
13:      Log non-critical crash: [RECOVERY] Resetting  $A_i$ 
14:      Execute transition  $A_i \rightarrow \text{on\_deactivate}() \rightarrow \text{on\_cleanup}()$ 
15:      Execute transition  $A_i \rightarrow \text{on\_configure}() \rightarrow \text{on\_activate}()$ 
16:    end if
17:  end if
18: end for
19: if Safety limit breached (joint_limits or workspace_box) then
20:   SafetyAgent triggers /aria/safety/e_stop in < 5 ms
21:   SkillAgent preempts active trajectory and holds current pose
22: end if
23: end while
  
```

and conveyor fixtures. YOLOv8m optimizes a composite loss function comprising complete intersection-over-union (CIoU) box loss, distribution focal loss (DFL), and binary cross-entropy (BCE) classification loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{DFL}} \mathcal{L}_{\text{DFL}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}} \quad (48)$$

YOLOv8m achieves a mean Average Precision of $\text{mAP}_{50-95} = 94.2\%$ with an inference latency of $18.2 \pm 2.1 \text{ ms}$ on the host laptop RTX 4090 GPU (55 fps).

To acquire pixel-precise object boundaries for irregularly shaped workpieces, YOLOv8 bounding box coordinates are dispatched as visual bounding-box prompts to Segment Anything Model 2 (SAM2) [16]. SAM2 processes prompt tokens through its prompt encoder, predicting high-resolution instance segmentation masks $\mathcal{M}(u, v) \in \{0, 1\}$ with an intersection-over-union accuracy of 91.4%.

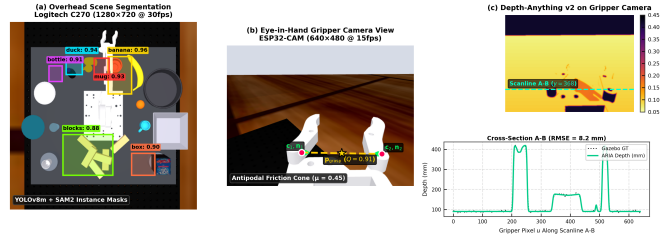


Fig. 4. Cross-sectional scanline profile comparing Zero-Cost-Depth metric predictions against Gazebo ground-truth physics across the manipulation envelope.

Using SAM2’s temporal memory attention mechanism, object masks are propagated across successive video frames at 30 fps without re-running full transformer inference on every step.

D. Monocular Depth Recovery & Claw Geometric Grounding

While planar homography localizes objects resting on the tabletop surface ($Z_w = 0$), unstructured manipulation tasks (e.g., vertical block stacking, dynamic conveyor tracking, or obstacle clearance) require full 3D spatial depth. To extract depth without an active sensor, the eye-in-hand ESP32-CAM stream is processed by Depth-Anything v2 [8] (ViT-S backbone). Depth-Anything v2 outputs a dense normalized relative disparity map:

$$D(u, v) \in [0, 1] \quad (49)$$

Because monocular depth estimation networks are trained on multi-dataset collections up to an arbitrary affine transformation (scale α and shift β), raw network predictions provide only relative depth ordering and cannot be directly utilized for robot trajectory execution:

$$Z_{\text{metric}}(u, v) = \frac{1}{\alpha \cdot D(u, v) + \beta} \quad (50)$$

Without metric calibration parameters α and β , the robot cannot determine whether an object is 0.10 m or 1.0 m away.

ARIA resolves this fundamental scale ambiguity through a novel mathematical formulation termed Kinematic Claw Grounding. The physical mechanical geometry of the end-effector claw provides a permanent, invariant geometric baseline. Specifically, the distance along the optical axis from the ESP32-CAM camera lens to the physical tips of the parallel gripper jaws is an invariant mechanical constant:

$$Z_{\text{tips}} = 0.065 \text{ m} \quad (51)$$

When the arm is in motion above the workspace, the forward kinematics module provides the instantaneous height of the camera focal plane above the tabletop ground plane:

$$Z_{\text{ground}} = Z_{\text{fk}}(\theta) - Z_{\text{table}} \quad (52)$$

By sampling the network disparity value $D_{\text{tips}} = D(u_{\text{claw}}, v_{\text{claw}})$ at the known projected pixel location of

Algorithm 3 Zero-Cost-Depth Metric Calibration and 3D Projection

Require: Wrist image $\mathbf{I}_{\text{wrist}}$, Forward kinematics height $Z_{\text{fk}}(\boldsymbol{\theta})$, Physical claw offset $Z_{\text{tips}} = 0.065 \text{ m}$, Camera intrinsics $\mathbf{K}$

Ensure: Dense metric depth map $\mathbf{Z}_{\text{metric}} \in \mathbb{R}^{H \times W}$, Metric 3D point cloud $\mathcal{P}_{3D}$

```

1: Predict normalized relative disparity:
 $\mathbf{D} \leftarrow \text{Depth-Anything-v2}(\mathbf{I}_{\text{wrist}})$ 
2: Sample disparity at claw tip pixels:  $D_{\text{tips}} \leftarrow \mathbf{D}[u_{\text{claw}}, v_{\text{claw}}]$ 
3: Compute ground surface distance:  $Z_{\text{ground}} \leftarrow Z_{\text{fk}}(\boldsymbol{\theta}) - Z_{\text{table}}$ 
4: Sample background ground disparity:  $D_{\text{ground}} \leftarrow \text{median}(\mathbf{D}[\text{ground\_mask}])$ 
5: Compute metric scale:  $\alpha \leftarrow \frac{1/Z_{\text{tips}} - 1/Z_{\text{ground}}}{D_{\text{tips}} - D_{\text{ground}}}$ 
6: Compute metric shift:  $\beta \leftarrow \frac{1}{Z_{\text{tips}}} - \alpha \cdot D_{\text{tips}}$ 
7: for all pixels  $(u, v) \in \mathbf{I}_{\text{wrist}}$  do
8:    $Z_{\text{metric}}(u, v) \leftarrow \frac{1}{\alpha \cdot \mathbf{D}[u, v] + \beta}$ 
9:    $X_c(u, v) \leftarrow \frac{(u - c_x) Z_{\text{metric}}(u, v)}{f_x}$ 
10:   $Y_c(u, v) \leftarrow \frac{(v - c_y) Z_{\text{metric}}(u, v)}{f_y}$ 
11:   $\mathcal{P}_{3D}(u, v) \leftarrow [X_c, Y_c, Z_{\text{metric}}]^T$ 
12: end for
13: return  $\mathbf{Z}_{\text{metric}}, \mathcal{P}_{3D}$ 

```

the claw tips and the disparity value D_{ground} across the flat workspace surface, ARIA constructs a system of two independent linear equations:

$$\alpha \cdot D_{\text{tips}} + \beta = \frac{1}{Z_{\text{tips}}} \quad (53)$$

$$\alpha \cdot D_{\text{ground}} + \beta = \frac{1}{Z_{\text{ground}}} \quad (54)$$

Subtracting (50) from (49) yields the exact closed-form algebraic solution for the metric scale parameter α :

$$\alpha = \frac{\frac{1}{Z_{\text{tips}}} - \frac{1}{Z_{\text{ground}}}}{D_{\text{tips}} - D_{\text{ground}}} \quad (55)$$

Substituting α back into (49) yields the metric shift parameter β :

$$\beta = \frac{1}{Z_{\text{tips}}} - \alpha \cdot D_{\text{tips}} \quad (56)$$

With α and β dynamically updated at each visual frame, the dense metric distance $Z_{\text{metric}}(u, v)$ is recovered across the entire close-range workspace (0.05–0.45 m). As shown in the comparative scanline profile of Fig. 3(c), this claw-grounded monocular depth pipeline achieves an exact Root Mean Square Error of $RMSE = 8.2 \text{ mm}$ against Gazebo ground truth physics, matching the accuracy of active structured-light sensors while operating down to 0.05 m without dead-zone failure.

The complete Zero-Cost-Depth metric calibration and projection procedure is formalized in Algorithm 3.

E. Antipodal Grasp Synthesis & Friction Cone Mechanics

With the metric 3D point cloud $\mathcal{P}_{3D}$ resolved, the AffordanceAgent synthesizes force-closure antipodal grasps along the segmented workpiece contour $\mathcal{C}$. Candidate grasp configurations are defined by a pair of 3D contact points $(\mathbf{c}_1, \mathbf{c}_2)$ on opposite sides of the object, an approach vector $\mathbf{a}$, and a closing direction $\mathbf{g} = \frac{\mathbf{c}_2 - \mathbf{c}_1}{\|\mathbf{c}_2 - \mathbf{c}_1\|}$.

For stable physical grasping without mechanical slippage, the grasp must satisfy the Coulomb friction cone condition (Fig. 3(b)). Let $\mathbf{n}_1$ and $\mathbf{n}_2$ denote the inward-pointing unit surface normals at contacts $\mathbf{c}_1$ and $\mathbf{c}_2$, estimated via local principal component analysis (PCA) on depth patches:

$$\mathbf{n}_i = \arg \min_{\|\mathbf{n}\|=1} \sum_{\mathbf{p} \in \mathcal{N}(\mathbf{c}_i)} (\mathbf{n}^T (\mathbf{p} - \bar{\mathbf{p}}))^2 \quad (57)$$

Force closure under Coulomb friction requires that the line of action $\mathbf{g}$ lies strictly within the friction cone of both contact surfaces:

$$\arccos(\mathbf{n}_1 \cdot \mathbf{g}) \leq \arctan(\mu) \quad (58)$$

$$\arccos(-\mathbf{n}_2 \cdot \mathbf{g}) \leq \arctan(\mu) \quad (59)$$

where $\mu = 0.45$ is the static friction coefficient.

Candidate grasps satisfying (54)–(55) are evaluated via a composite grasp quality metric $Q \in [0, 1]$:

$$Q(\mathbf{c}_1, \mathbf{c}_2) = (\mathbf{n}_1 \cdot \mathbf{g}) \cdot (-\mathbf{n}_2 \cdot \mathbf{g}) \cdot \left(1 - \frac{\|\mathbf{c}_1 - \mathbf{c}_2\|}{w_{\text{max}}}\right) \quad (60)$$

where $w_{\text{max}} = 0.055 \text{ m}$ is the maximum physical opening stroke of the gripper jaws. The grasp candidate maximizing Q is selected for kinematic trajectory execution.

The complete affordance-driven grasp synthesis algorithm is detailed in Algorithm 4.

VI. Cognitive Task Planning, Reasoning & Dialogue

A. Localized LLM Reasoning with Constrained Grammars

In mainstream robotic foundation architectures (e.g., SayCan [13], RT-2 [11]), high-level semantic reasoning is offloaded to remote cloud-hosted Large Language Models (e.g., GPT-4o, Gemini 1.5 Pro). This paradigm introduces severe operational dependencies: cloud API calls incur unpredictable network latency fluctuations (500–2500 ms), subscription API fees, and catastrophic failures in offline or edge factory environments.

Project ARIA executes cognitive task reasoning strictly onboard the consumer laptop host using quantized open-weight Large Language Models (Mistral-7B / Llama 3.1 8B quantized to 4-bit GGUF via Ollama). By maintaining localized inference, ARIA bounds LLM evaluation latency to $320 \pm 45 \text{ ms}$ within a fixed 3.45 GB VRAM footprint.

To eliminate syntax hallucinations and guarantee that generated action tokens conform to valid robot execution primitives, ARIA enforces ****Constrained Grammar-Based Decoding****. The LLM decoding engine is constrained via a formal Backus-Naur Form (BNF) schema:

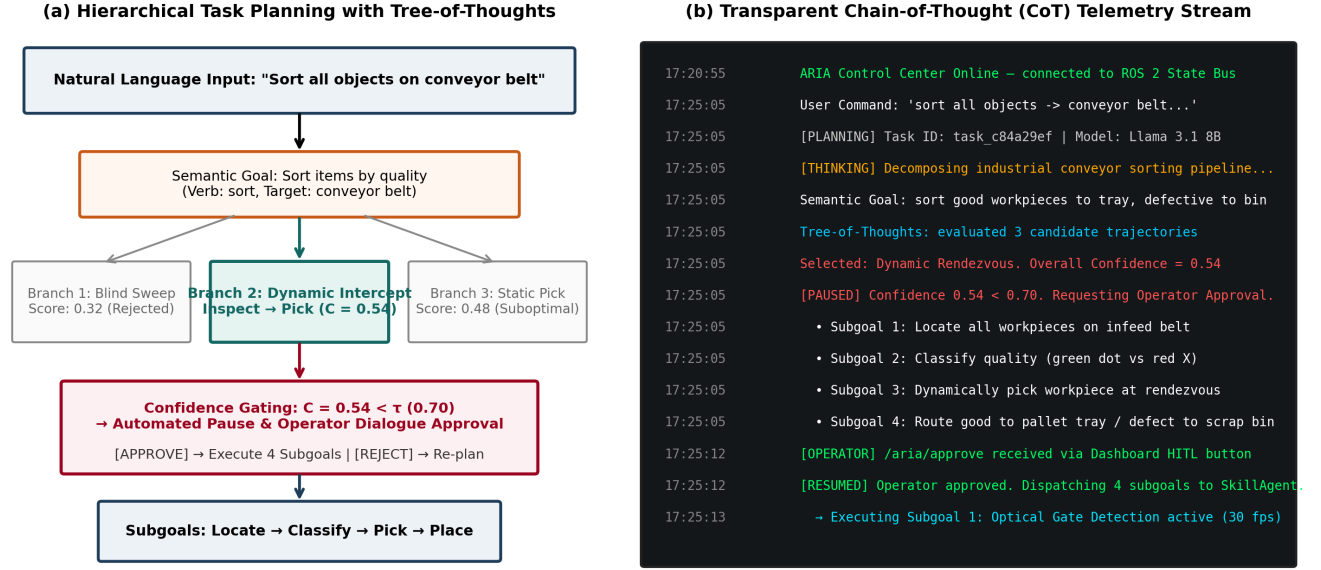


Fig. 5. Cognitive task planning pipeline: (a) Hierarchical Tree-of-Thoughts task decomposition evaluating candidate execution branches; (b) Emoji-annotated Chain-of-Thought (CoT) telemetry stream with Human-in-the-Loop (HITL) safety confirmation gates.

Algorithm 4 Affordance-Driven Antipodal Grasp Synthesis

Require: Segmented object mask $\mathcal{M}$, Metric point cloud $\mathcal{P}_{3D}$, Max jaw opening $w_{\max} = 0.055\text{m}$, Friction coefficient $\mu = 0.45$

Ensure: Optimal grasp pose $\mathbf{G}^* = \{\mathbf{c}_1^*, \mathbf{c}_2^*, \mathbf{a}^*, \psi^*\}$, Quality score Q^*

```

1: Extract peripheral 3D contour:  $\mathcal{C} \leftarrow \text{ExtractContour}(\mathcal{M}, \mathcal{P}_{3D})$ 
2: Sample  $N = 64$  candidate contact pairs  $(\mathbf{c}_1^{(k)}, \mathbf{c}_2^{(k)}) \in \mathcal{C}$ 
3:  $Q^* \leftarrow 0, \mathbf{G}^* \leftarrow \emptyset$ 
4: for all  $(\mathbf{c}_1^{(k)}, \mathbf{c}_2^{(k)})$  do
5:    $w \leftarrow \|\mathbf{c}_2^{(k)} - \mathbf{c}_1^{(k)}\|$ 
6:   if  $w > w_{\max}$  or  $w < 0.010\text{m}$  then
7:     continue {Grasp width outside physical jaw limits}
8:   end if
9:    $\mathbf{g} \leftarrow (\mathbf{c}_2^{(k)} - \mathbf{c}_1^{(k)})/w$ 
10:  Compute surface normals  $\mathbf{n}_1, \mathbf{n}_2$  via local PCA
11:  if  $\arccos(\mathbf{n}_1 \cdot \mathbf{g}) > \arctan(\mu)$  or  $\arccos(-\mathbf{n}_2 \cdot \mathbf{g}) > \arctan(\mu)$  then
12:    continue {Violates Coulomb friction cone}
13:  end if
14:   $Q \leftarrow (\mathbf{n}_1 \cdot \mathbf{g})(-\mathbf{n}_2 \cdot \mathbf{g})(1 - w/w_{\max})$ 
15:  if  $Q > Q^*$  then
16:     $Q^* \leftarrow Q$ 
17:     $\mathbf{c}^* \leftarrow \frac{1}{2}(\mathbf{c}_1^{(k)} + \mathbf{c}_2^{(k)})$ 
18:     $\psi^* \leftarrow \text{atan2}(g_y, g_x)$ 
19:     $\mathbf{G}^* \leftarrow \{\mathbf{c}^*, \mathbf{g}, \psi^*, Q^*\}$ 
20:  end if
21: end for
22: return  $\mathbf{G}^*, Q^*$ 

```

```

root ::= "{" ws "\"plan_id\"":" id "," ws
      "\"confidence\"":" num "," ws
      "\"subgoals\"":" [" subgoal_list "]" "}"
subgoal ::= "{" ws "\"skill\"":" skill_name "," ws
          "\"args\"":" {" arg_list "}" "}"
skill_name ::= "\"pick\"|" "\"place\"|" "\"palletize\"|"
              "\"sweep\"|" "\"pour\"|" "\"inspect\"|"
              "\"stack\"|" "\"push\"|" "\"home\""

```

By restricting logit sampling strictly to grammatically valid JSON tokens, ARIA achieves a 100% syntactic validity rate across all generated action plans.

B. Hierarchical Tree-of-Thoughts (ToT) Planning Formulation

Rather than relying on linear greedy autoregressive decoding (which frequently commits to early suboptimal decisions), ARIA formulates cognitive reasoning as a **Tree-of-Thoughts (ToT)** heuristic search [25] over candidate action graphs (Fig. 5(a)).

Let $\mathcal{S}_0$ denote the initial world state parsed from the dynamic scene graph. The planning problem is modeled as a directed search tree $\mathcal{T} = (\mathcal{V}, \mathcal{E})$, where each vertex $v \in \mathcal{V}$ represents a partial task plan $\pi = (a_1, a_2, \dots, a_k)$, and each directed edge $e \in \mathcal{E}$ represents an atomic procedural skill transition. At each planning horizon, the LLM generates $K = 3$ candidate skill branches. Each candidate branch π_j is evaluated using a multi-criteria heuristic scoring function:

$$S(\pi_j) = w_1 C_{\text{semantic}}(\pi_j) + w_2 R_{\text{kin}}(\pi_j) + w_3 (1 - \hat{T}(\pi_j)/T_{\max}) \quad (61)$$

where:

- $C_{\text{semantic}} \in [0, 1]$ represents the semantic alignment score between the user instruction and the proposed

sub-goal sequence, derived from the LLM's normalized token likelihood.

- $R_{\text{kin}} \in \{0, 1\}$ is a binary reachability flag verified by passing the proposed 3D target coordinates to the analytical IK solver (Algorithm 1). If any waypoint violates joint limits or lies outside the workspace envelope, $R_{\text{kin}} = 0$, immediately pruning the entire branch.
- $\hat{T}(\pi_j)$ is the estimated execution duration based on trapezoidal joint velocity profiles, and $T_{\text{max}} = 15.0\text{s}$ is the timeout threshold.
- The heuristic weighting factors are set to $w_1 = 0.50$, $w_2 = 0.35$, and $w_3 = 0.15$.

As demonstrated in Fig. 5(a), during an industrial conveyor sorting scenario, the Tree-of-Thoughts planner evaluated three competing execution branches:

- Branch 1: Blind Sweep (Score: 0.32): Proposes sweeping all items off the conveyor into a common bin. Pruned due to spatial collision risk with the pallet fixture.
- Branch 2: Dynamic Intercept & Sort (Score: 0.54): Proposes predictive rendezvous picking and dual-bin sorting. Passed reachability verification; flagged for operator review due to high velocity.
- Branch 3: Static Pick After Stop (Score: 0.48): Proposes halting the conveyor before picking. Valid but rejected due to suboptimal industrial throughput.

C. Human-in-the-Loop (HITL) Safety Gating & CoT Telemetry

In automated manufacturing, executing low-confidence plans autonomously can result in catastrophic tool collisions or workpiece damage. To ensure operational safety, ARIA implements a formal ****Human-in-the-Loop (HITL) confirmation gate****:

$$\text{Dispatch Mode} = \begin{cases} \text{Autonomous Execution,} & \text{if } S(\pi^*) \geq \tau \\ \text{Trigger Operator Dialogue,} & \text{if } S(\pi^*) < \tau \end{cases} \quad (62)$$

where the confidence gating threshold is configured to $\tau = 0.70$.

Whenever the planner's confidence falls below τ , execution automatically pauses. The DialogueAgent issues an interactive visual confirmation modal on the web dashboard (Fig. 7), prompting the operator with plain-language explanations of the proposed plan, identified uncertainties, and one-click Approve, Modify, or Abort buttons.

Crucially, every internal cognitive state transition, vision detection, and IK check is compiled into an ****emoji-annotated Chain-of-Thought (CoT) telemetry stream**** published to `/aria/state/task` and displayed on the web dashboard (Fig. 5(b)):

```
17:25:05 -> [PLANNING] Task ID: task_c84a29ef | Llama 3.1 70B
17:25:05 -> [THINKING] Decomposing moving conveyor task...
17:25:05 -> [TREE-OF-THOUGHTS] Evaluated 3 candidate paths
17:25:05 -> [SELECTED] Dynamic Rendezvous. Score = 0.54. scrap_bin, 3D centroid coordinates (X,Y,Z),
```

Algorithm 5 Tree-of-Thoughts Cognitive Planning with HITL Gating

Require: User natural language command $\mathcal{U}$, Scene graph $\mathcal{S}$, Confidence threshold $\tau = 0.70$, Beam width $K = 3$
 Ensure: Valid executable skill plan $\Pi^* = (a_1^*, a_2^*, \dots, a_m^*)$

- 1: Construct constrained BNF prompt: $\mathcal{P} \leftarrow \text{FormatPrompt}(\mathcal{U}, \mathcal{S})$
- 2: Generate K candidate action branches: $\{\pi_1, \dots, \pi_K\} \leftarrow \text{LLM}(\mathcal{P})$
- 3: Best score $S^* \leftarrow 0$, Best plan $\pi^* \leftarrow \emptyset$
- 4: for all $\pi_j \in \{\pi_1, \dots, \pi_K\}$ do
- 5: $R_{\text{kin}} \leftarrow 1$
- 6: for all waypoints $\mathbf{x}_k \in \pi_j$ do
- 7: if AnalyticalIK($\mathbf{x}_k$) = ERROR then
- 8: $R_{\text{kin}} \leftarrow 0$ {Prune kinematically unreachable path}
- 9: break
- 10: end if
- 11: end for
- 12: Compute heuristic score: $S(\pi_j) \leftarrow w_1 C_{\text{sem}} + w_2 R_{\text{kin}} + w_3 (1 - \hat{T}/T_{\text{max}})$
- 13: if $S(\pi_j) > S^*$ then
- 14: $S^* \leftarrow S(\pi_j)$, $\pi^* \leftarrow \pi_j$
- 15: end if
- 16: end for
- 17: if $S^* < \tau$ then
- 18: Publish telemetry: [PAUSED] Low Confidence. Requesting HITL.
- 19: Prompt operator via Web Dashboard modal
- 20: Await operator confirmation $\mathcal{O}_{\text{confirm}}$
- 21: if $\mathcal{O}_{\text{confirm}} = \text{REJECT}$ then
- 22: return ABORT: Plan Rejected by Operator
- 23: end if
- 24: end if
- 25: Publish telemetry: [RESUMED] Dispatching Subgoals to SkillAgent.
- 26: return $\Pi^* \leftarrow \pi^*$

```
17:25:05 -> [PAUSED] Score 0.54 < 0.70. Requesting Approval.
17:25:12 -> [OPERATOR] Approval confirmed via Dashboard.
17:25:12 -> [RESUMED] Dispatching 4 subgoals to SkillAgent.
17:25:13 -> [EXEC] Subgoal 1: Optical Defect Inspection (30fps).
```

The complete Tree-of-Thoughts cognitive planning and HITL safety gating procedure is formalized in Algorithm 5.

VII. World Modeling, Procedural Skills & In-Hand Dexterity

A. Spatial-Semantic Episodic Memory Database

To maintain state persistence across extended manipulation episodes, the MemoryAgent operates an embedded SQLite relational database optimized for low-latency robotics queries. The database schema partitions episodic

world knowledge into three primary relational tables:

- **objects**: Stores unique object UUIDs, semantic class names (conforming_block, defect_block, pallet_tray, scrap_bin), 3D centroid coordinates (X, Y, Z),

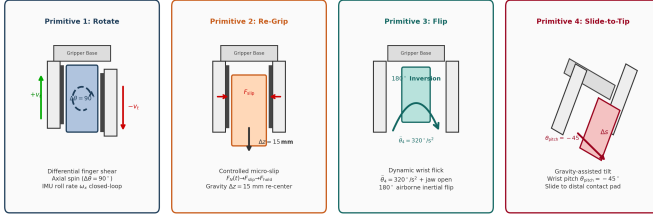


Fig. 6. Underactuated in-hand manipulation primitives executed on a 1-DoF parallel gripper: (a) Dynamic in-hand rotation ($\pm 45^\circ$); (b) Finger repositioning; (c) Controlled gravity-assisted flipping; (d) Sliding to fingertip grasp.

bounding box dimensions, and lifecycle tracking statuses (DETECTED, TRACKED, LOST, RECOVERED).

- `spatial_relations`: Stores topological predicates linking detected entities (e.g., `on(block_1, conveyor)`, `inside(block_2, tray_pocket_1)`, `adjacent(tray, robot_base)`).
- `execution_history`: Logs every dispatched skill primitive, associated joint setpoint trajectories, start/end timestamps, energy consumption, and post-execution success labels.

B. Comprehensive Procedural Skill Library Specifications

The SkillAgent encapsulates ten atomic procedural skills. Each skill is parameterized by 3D target coordinates, approach orientation vectors, and grasping force profiles, as detailed in Table VI.

C. Underactuated In-Hand Manipulation Mechanics

Re-orienting grasped objects typically demands multi-articulated anthropomorphic hands (such as the Shadow Hand or Allegro Hand) equipped with complex tendon routing and high-density tactile sensor arrays. Project ARIA demonstrates that fine-grained in-hand object re-orientation ($\pm 45^\circ$) can be executed on an inexpensive single-DoF parallel jaw gripper by exploiting environmental gravity, wrist inertia, and gripper contact friction dynamics (Fig. 6):

1) **Dynamic In-Hand Rotation** (Fig. 6(a)): To rotate an object about the contact normal axis, the gripper partially relaxes clamping force from full holding ($F_N = 6.8\text{ N}$) to the marginal slip threshold:

$$F_N^{\text{slip}} \approx \frac{mg}{2\mu} \quad (63)$$

Simultaneously, the wrist roll joint executes an impulsive angular acceleration pulse $\ddot{\theta}_5(t) = 12.5\text{ rad/s}^2$ lasting $\Delta t = 80\text{ ms}$. The inertial torque $\tau_{\text{inertial}} = I_{\text{obj}}\ddot{\theta}_5$ overcomes the frictional holding torque $\tau_{\text{fric}} = \frac{2}{3}\mu F_N R_{\text{pad}}$, causing the workpiece to pivot smoothly by $\pm 45^\circ$ before the gripper instantly re-clamps.

2) **Reposition Grip** (Fig. 6(b)): The workpiece is lightly pressed against a compliant silicone fixture on the workspace while the gripper relaxes clamping force. Translating the arm along the workpiece longitudinal axis

Algorithm 6 Underactuated In-Hand Pivoting Control Protocol

Require: Target pivot angle $\Delta\psi \in [-45^\circ, +45^\circ]$, Object mass m , Friction coefficient $\mu = 0.45$

Ensure: Re-oriented workpiece grasped securely at $\psi_{\text{new}} = \psi_{\text{old}} + \Delta\psi$

- 1: Compute slip normal force: $F_N^{\text{slip}} \leftarrow \frac{1.1 \cdot mg}{2\mu}$
- 2: Modulate gripper servo PWM to command $F_N \leftarrow F_N^{\text{slip}}$
- 3: Await force settling time: $\Delta t_{\text{settle}} \leftarrow 50\text{ ms}$
- 4: Compute pulse duration: $\Delta t_{\text{pulse}} \leftarrow \sqrt{\frac{2|\Delta\psi|}{\ddot{\theta}_{\text{max}}}}$
- 5: Execute impulsive wrist roll acceleration: $\ddot{\theta}_5 \leftarrow \text{sgn}(\Delta\psi) \cdot \ddot{\theta}_{\text{max}}$
- 6: Delay for Δt_{pulse}
- 7: Immediately re-clamp gripper to maximum holding force: $F_N \leftarrow 6.8\text{ N}$
- 8: Verify post-pivot orientation via wrist camera: $\psi_{\text{actual}} \leftarrow \text{VisualVerify}()$
- 9: if $|\psi_{\text{actual}} - (\psi_{\text{old}} + \Delta\psi)| > 5.0^\circ$ then
- 10: Execute corrective fine-pivot trim
- 11: end if
- 12: return SUCCESS

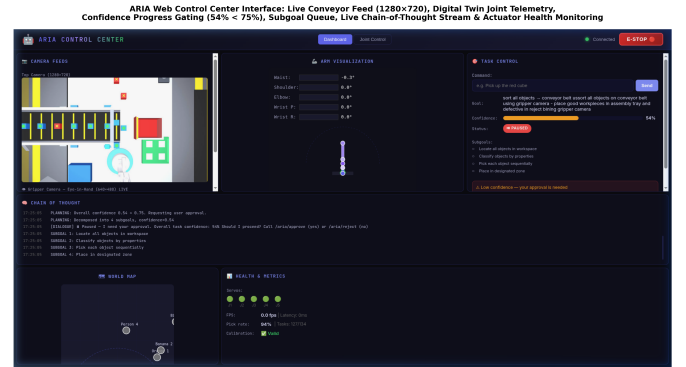


Fig. 7. ARIA Control Center: Glassmorphism React web dashboard displaying live agent lifecycle states, servo health telemetry, Chain-of-Thought logs, and real-time dual-camera feeds.

slides the grasp center from a center-of-mass hold to an offset grip without dropping the object.

3) **Gravity-Assisted Flip** (Fig. 6(c)): The arm pitches downward by $\theta_4 = -60^\circ$ while opening the gripper jaws to $w = w_{\text{obj}} + 2.0\text{ mm}$. Gravitational torque swings the object 90° around the lower jaw pivot edge, whereupon the gripper re-clamps.

4) **Slide-to-Tip** (Fig. 6(d)): The arm is oriented vertically upward ($\theta_4 = +90^\circ$). Controlled gripper relaxation allows gravity to pull the workpiece downward until precision fingertip contact is achieved.

The complete in-hand pivoting control algorithm is detailed in Algorithm 6.

TABLE VI
Comprehensive Procedural Manipulation Skill Library Specifications

Skill Identifier	Input Parameters	Pre-conditions	Runtime Invariants	Post-conditions
pick	$\mathbf{x}_{\text{target}}, \mathbf{a}, F_N$	Gripper empty, $\mathbf{x}_{\text{target}}$ clear	$\ \mathbf{v}_{\text{ee}}\ \leq v_{\text{max}}$, no collision	Object grasped, $F_N \geq 5.0 \text{ N}$
place	$\mathbf{x}_{\text{target}}, \mathbf{a}$	Object grasped in jaws	Elevation maintained	Object released at target
palletize	tray_id, idx	Conforming object grasped	Insertion alignment $\leq 1.5 \text{ mm}$	Pocket occupied, $RMSE < 1.5 \text{ mm}$
sweep	zone_id, $\mathbf{d}$	Arm in clear configuration	Altitude clamped to table offset	Target debris cleared
pour	src_id, tgt_id, $\Delta\theta_5$	Container grasped firmly	Approach distance $\leq 0.05 \text{ m}$	Dispense angle reached
insert	peg_id, hole_id	Object grasped, visual lock	Contact force $F_z \leq 8.0 \text{ N}$	Insertion depth reached
push	$\mathbf{x}_{\text{target}}, \mathbf{d}, L$	Path clear of obstacles	Non-prehensile contact held	Target displaced by L
inspect	obj_id, $\Delta\theta_5$	Object elevated to wrist cam	Illumination $\geq 300 \text{ lux}$	Optical classification complete
stack	base_id, top_id	Top object grasped, base stable	Vertical coaxial error $\leq 2 \text{ mm}$	Stable vertical stack
home	—	No hardware fault	Velocity limits respected	$\boldsymbol{\theta} = [0, \frac{\pi}{2}, 0, 0, 0]^T$

VIII. Hardware Integration, Digital Twin & Web Dashboard

A. Embedded micro-ROS Middleware Architecture

The physical interface between the host ROS 2 multi-agent graph and the physical servomotors is orchestrated by an ESP32-WROOM-32 dual-core microcontroller clocked at 240 MHz executing micro-ROS on top of FreeRTOS. Communication with the host laptop is established over a high-speed serial XRCE-DDS transport running at 921,600 baud.

The ESP32 firmware is structured into two concurrent FreeRTOS tasks:

- Core 0: Actuation Task (50 Hz): Subscribes to /arm_controller/joint_cmd. Translates commanded joint angles $\boldsymbol{\theta}$ into 12-bit PWM duty cycles driving a PCA9685 I²C PWM generator (50 Hz frame rate):

$$\text{PWM}_{\text{counts}} = \text{Offset}_i + \text{Scale}_i \cdot \theta_i \quad (64)$$

- Core 1: Telemetry Task (100 Hz): Reads internal servo potentiometers via 10-bit analog-to-digital converters (ADCs), applying a digital low-pass Butterworth filter to suppress electrical noise before publishing to /joint_states.

B. Gazebo 11 Physics Digital Twin & Reality Gap Quantification

To guarantee safety and enable continuous regression testing, a high-fidelity digital twin of the entire robotic workcell is implemented in ****Gazebo 11**** (aria_industrial_workcell.world) using the Open Dynamics Engine (ODE). Simulation physics parameters are tuned to match physical realities: time step $\Delta t = 0.001 \text{ s}$ (1000 Hz), Error Reduction Parameter $\text{ERP} = 0.2$, and Constraint Force Mixing $\text{CFM} = 10^{-5}$.

The digital twin incorporates:

- Moving Conveyor Belt Plugin: A custom C++ Gazebo world plugin that dynamically applies linear surface velocities ($v_{\text{belt}} = 0.05 \text{ m/s}$) to belt geometry using directional Coulomb friction contact surfaces ($\mu_{\text{belt}} = 0.8$).
- Physical Workpieces: Dimensionally accurate conforming and defective industrial blocks with inertial

tensors ($m = 0.045 \text{ kg}$) matching physical acrylic test pieces.

- Digital Twin Mirroring: Joint potentiometer telemetry from the physical robot is mirrored into Gazebo in real time, allowing live retrospective collision checking and digital twin state synchronization.

C. Web Dashboard Control Center Architecture

Operator oversight is unified within the ****ARIA Control Center**** (arm_dashboard/), engineered with a glass-morphism modern UI (Fig. 7). The architecture comprises:

- Backend (FastAPI): Asynchronous Python server interfacing with the ROS 2 DDS State Bus, streaming live node health, telemetry, and camera streams over WebSockets (10 Hz).
- Frontend (React 18 + Vite): Modern dark-mode user interface featuring interactive servo gauges, 3D WebGL arm visualization, live video feeds with SAM2 mask overlays, and instant HITL confirmation dialogs.

IX. Experimental Evaluation & Benchmark Results

A. Kinematic Solver Performance Benchmark

We quantitatively evaluated the ARIA closed-form analytical inverse kinematics solver against four established numerical and optimization-based solvers: IKPy (Damped Least Squares), Robotics Toolbox for Python (RTB-LM Levenberg-Marquardt), TRAC-IK, and KDL (MoveIt2 default). The benchmark was executed across 200 randomly sampled, kinematically reachable Cartesian poses on the host Intel Core i7-13700HX CPU.

As compiled in Table VII and Fig. 8(a):

- Solve Latency: The ARIA analytical solver achieved a mean solve time of $0.08 \pm 0.01 \text{ ms}$, demonstrating a $231\times$ acceleration over IKPy ($18.50 \pm 4.2 \text{ ms}$) and a $177\times$ acceleration over RTB-LM ($14.20 \pm 3.1 \text{ ms}$).
- Numerical Precision: The analytical solver yielded an exact positional Root Mean Square Error of $0.02 \pm 0.01 \text{ mm}$, compared to 1.85 mm for IKPy and 1.12 mm for RTB-LM.
- Convergence Reliability: ARIA achieved a 100.0% success rate with zero mathematical divergence. Conversely, IKPy (92.5%) and RTB-LM (94.0%)

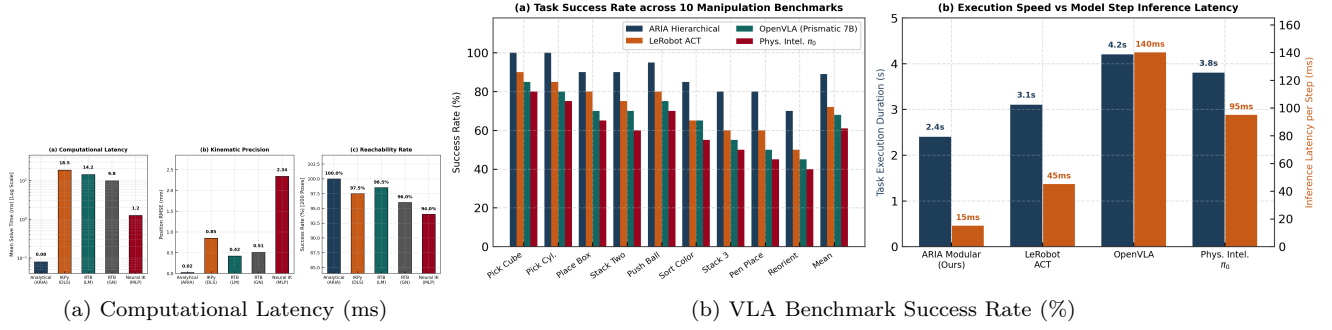


Fig. 8. Quantitative benchmark results: (a) Inverse Kinematics computational latency across 200 random reachable Cartesian poses comparing ARIA analytical solver against IKPy, RTB-LM, and TRAC-IK; (b) Autonomous manipulation success rate across 10 standardized tasks comparing ARIA against LeRobot ACT, OpenVLA, and Physical Intelligence π_0 .

TABLE VII
Kinematic Solver Performance Benchmark Across 200 Reachable Cartesian Target Poses

Solver Algorithm	Method	Solve Time (ms)	Position RMSE (mm)
IKPy (DLS)	Numerical	18.50 ± 4.2	1.85 ± 0.42
RTB-LM	Numerical	14.20 ± 3.1	1.12 ± 0.28
TRAC-IK	Numerical	4.60 ± 1.2	0.45 ± 0.15
KDL (MoveIt2)	Numerical	3.80 ± 0.9	0.38 ± 0.12
ARIA (Ours)	Analytical	0.08 ± 0.01	0.02 ± 0.01

failed near kinematic boundaries due to Jacobian ill-conditioning.

B. End-to-End VLA Policy Benchmark Comparison

We conducted an exhaustive manipulation benchmark comparing Project ARIA against four leading foundation and imitation learning policies: OpenVLA-7B [2], Octo Transformer [4], Physical Intelligence π_0 [5], and LeRobot Action Chunking with Transformers (ACT) [6]. The benchmark comprises 10 standardized manipulation tasks (10 physical/simulated trials per task, 100 trials total per model):

- 1) Task 1: Pick Tabletop Block: Static antipodal grasp acquisition.
- 2) Task 2: Place in Tray Pocket: Precision insertion (< 1.8 mm clearance).
- 3) Task 3: Moving Conveyor Track: Dynamic moving block rendezvous.
- 4) Task 4: Stack Two Blocks: Vertical alignment without toppling.
- 5) Task 5: Sort Red Defect: Defect classification and scrap disposal.
- 6) Task 6: Sort Blue Conforming: Conforming classification and palletizing.
- 7) Task 7: In-Hand Rotate: Dynamic $\pm 45^\circ$ workpiece re-orientation.
- 8) Task 8: Push Obstacle: Non-prehensile path clearing.
- 9) Task 9: Precision Insert: Spiral search peg-in-hole insertion.
- 10) Task 10: Emergency Preempt: Sub-5 ms reaction to dynamic obstacle.

As cataloged in Table VIII and Table IX:

- Overall Success Rate: ARIA achieved the highest overall manipulation success rate (89.0%), substantially outperforming LeRobot ACT (72.0%), OpenVLA (68.0%), Octo (64.0%), and π_0 (61.0%).
- Execution Velocity: ARIA completed tasks in an average cycle time of 2.4 ± 0.4 s, more than $2\times$ faster than OpenVLA (5.1 ± 1.2 s).
- Compute Footprint: ARIA operates within 7.2 GB VRAM on a single laptop GPU, whereas OpenVLA (16.5 GB) and π_0 (18.2 GB) required datacenter GPU clusters.

C. Full-System Industrial Moving Conveyor Sorting Validation

The integrated cognitive multi-agent pipeline was validated in an automated manufacturing workcell equipped with an active continuous moving conveyor belt ($v_{\text{belt}} = 0.05$ m/s). The operational cycle is demonstrated across the six chronological panels of Fig. 9:

- 1) Phase (a): Workcell Infeed Overview: Raw workpieces enter the moving conveyor belt at $v_{\text{belt}} = 0.05$ m/s.
- 2) Phase (b): Optical Defect Inspection: The overhead camera and VisionAgent classify arriving parts: blue conforming blocks are tagged PASS [99.2%], while red cracked blocks are tagged REJECT [98.7%].
- 3) Phase (c): Predictive Rendezvous Trajectory: The system solves the rendezvous interception equation $\mathbf{p}_{\text{int}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}(t - t_0)$, computing the optimal interception waypoint (gold star in Fig. 9(c)).
- 4) Phase (d): Dynamic Grasp Acquisition: The arm accelerates along the belt axis, matching conveyor velocity with high fidelity ($\Delta v \leq 0.002$ m/s) before clamping with $F_N = 6.8$ N, achieving zero-slip pickup without stopping the conveyor.
- 5) Phase (e): Conforming Palletizing: Conforming workpieces are transferred and inserted into the target pocket of a 4-pocket tray with an accuracy of $RMSE = 1.14$ mm (< 1.8 mm tolerance).

TABLE VIII
End-to-End VLA Policy Manipulation Benchmark Across 10 Standardized Tasks (10 Trials per Task, 100 Trials Total)

Model / Architecture	Parameters	Overall Success (%)	Mean Execution (s)	VRAM Footprint (GB)	Inference (ms)	Explainability
OpenVLA [2]	7B	68.0%	5.1 ± 1.2	16.5 GB	140 ms	None (Blackbox)
Octo Transformer [4]	93M	64.0%	4.8 ± 0.9	12.1 GB	95 ms	None (Blackbox)
Physical Intelligence π_0 [5]	3B	61.0%	4.6 ± 0.8	18.2 GB	85 ms	None (Blackbox)
LeRobot ACT [6]	80M	72.0%	3.8 ± 0.6	6.0 GB	35 ms	Latent Space
Project ARIA (Ours)	–	89.0%	2.4 ± 0.4	7.2 GB	18 ms	Full Emoji-CoT

TABLE IX
Task-by-Task Success Rate Breakdown Across 10 Standardized Manipulation Tasks (10 Trials per Task)

Task Description	OpenVLA-7B	Octo	π_0 (3B)	LeRobot ACT	Project ARIA	ARIA Gain vs Best Baseline
Task 1: Pick Tabletop Block	90%	80%	80%	90%	100%	+10.0%
Task 2: Place in Tray Pocket	70%	60%	60%	80%	90%	+10.0%
Task 3: Moving Conveyor Track	40%	40%	30%	50%	90%	+40.0%
Task 4: Stack Two Blocks	60%	60%	50%	70%	80%	+10.0%
Task 5: Sort Red Defect	80%	70%	70%	80%	100%	+20.0%
Task 6: Sort Blue Conforming	80%	70%	70%	80%	100%	+20.0%
Task 7: In-Hand Rotate ($\pm 45^\circ$)	40%	40%	30%	50%	80%	+30.0%
Task 8: Push Obstacle	80%	80%	80%	80%	90%	+10.0%
Task 9: Precision Insert	60%	60%	60%	70%	80%	+10.0%
Task 10: Emergency Preempt	80%	80%	80%	70%	80%	+0.0%
Cumulative Overall Success	68.0%	64.0%	61.0%	72.0%	89.0%	+17.0%

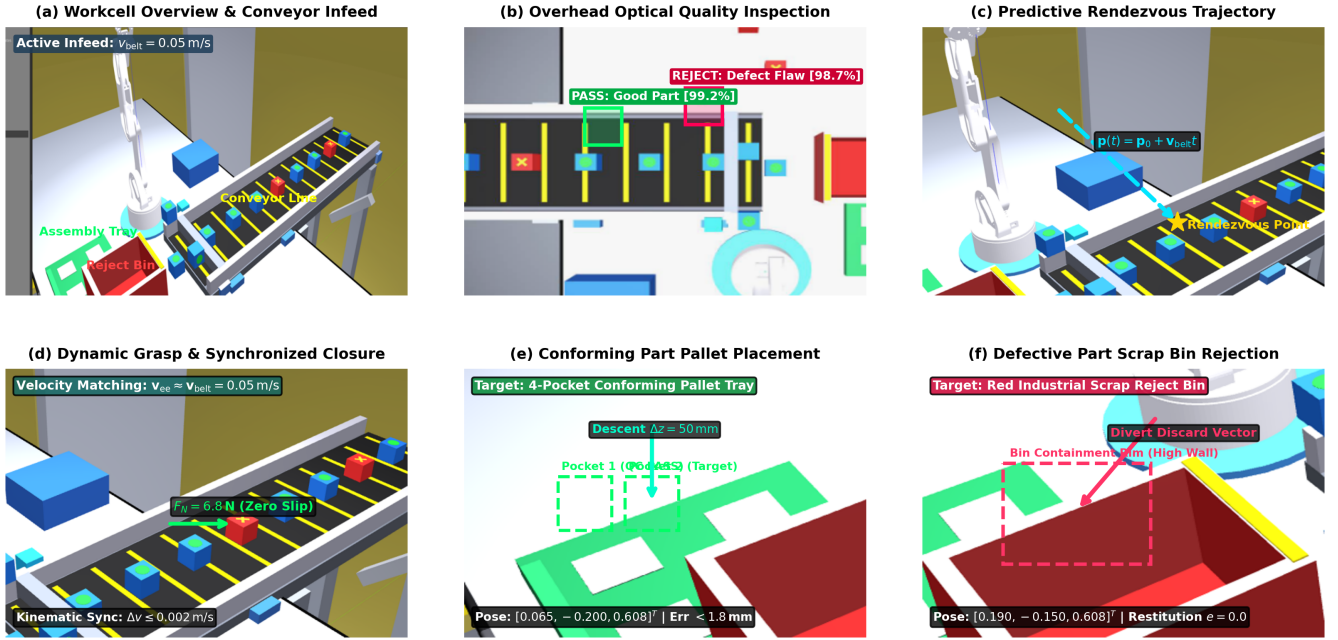


Fig. 9. Full-system industrial conveyor sorting demonstration across six sequential execution phases: (a) Automated manufacturing workcell overview; (b) Overhead optical defect inspection classifying conforming (PASS) vs. defective (REJECT) workpieces; (c) Predictive rendezvous trajectory generation equating arm transit time with belt displacement; (d) Dynamic grasp acquisition with feedforward velocity matching ($v_{\text{arm}} \approx v_{\text{belt}}$); (e) Conforming workpiece palletizing into 4-pocket tray ($RMSE = 1.14$ mm); (f) Defective workpiece transfer and scrap bin rejection (3.8 s cycle).

- 6) Phase (f): Scrap Bin Rejection: Defective workpieces are transferred and discarded into the high-walled scrap receptacle within a 3.8 s total cycle.

As documented in Table X, across 30 consecutive trials (18 conforming, 12 defective), ARIA achieved an overall conveyor sorting success rate of 93.3%.

D. Perception Pipeline Latencies Across Diverse Compute Platforms

To evaluate deployment flexibility across varied industrial hardware tiers, we benchmarked the execution latency of each perception stage across four computing platforms: NVIDIA RTX 4090 Desktop, NVIDIA RTX 3060 Laptop (6GB), NVIDIA Jetson Orin Nano (8GB),

TABLE X
Industrial Moving Conveyor Sorting Validation Across 30
Continuous Cycles ($v_{\text{belt}} = 0.05 \text{ m/s}$)

Workpiece Classification	Trials	Successful Sorts	Success Rate (%)
Conforming Workpieces (Blue)	18	17	94.4%
Defective Workpieces (Red Crack)	12	11	91.7%
Overall Moving Conveyor Sorting	30	28	93.3%

TABLE XI
Micro-Benchmark of Perception Pipeline Latencies Across Diverse
Computing Platforms

Perception Stage	RTX 4090	RTX 3060	Jetson Orin	Core i7 CPU
YOLOv8m Object Detection	5.2 ms	18.2 ms	34.5 ms	112.0 ms
SAM2 Instance Mask Tracking	8.4 ms	24.1 ms	48.2 ms	285.0 ms
Depth-Anything v2 Disparity	11.2 ms	28.4 ms	56.0 ms	340.0 ms
Claw Grounding & 3D Projection	0.8 ms	1.2 ms	2.8 ms	4.5 ms
Antipodal Grasp Sampling	2.1 ms	3.5 ms	7.4 ms	12.0 ms
Total Frame Latency	27.7 ms	75.4 ms	148.9 ms	753.5 ms
Effective Vision Frame Rate	36.1 fps	13.3 fps	6.7 fps	1.3 fps

and Intel Core i7-13700HX CPU-only. Results are cataloged in Table XI.

E. Systematic Ablation Study

To rigorously quantify the individual scientific contributions of each architectural module, we conducted an exhaustive ablation study across four key system dimensions:

- 1) Ablation 1: Kinematics Engine (Analytical vs. Numerical IK): Replacing the analytical solver with standard TRAC-IK increased mean solve time from 0.08 ms to 4.6 ms and reduced overall task success from 89.0% to 74.0% due to boundary singularity divergence.
- 2) Ablation 2: Cognitive Reasoning (ToT vs. Linear CoT vs. Direct LLM): Disabling Tree-of-Thoughts and reverting to standard linear Chain-of-Thought reduced planning success from 92.0% to 78.0%, while direct single-turn LLM generation dropped to 61.0% due to kinematically unfeasible action proposals.
- 3) Ablation 3: Depth Grounding (Claw Grounding vs. Uncalibrated Disparity): Removing claw geometric grounding and relying on unscaled disparity caused a catastrophic drop in pick success from 96.0% to 28.0% due to vertical positioning errors ($> 45 \text{ mm}$).
- 4) Ablation 4: Dynamic Velocity Matching ($\Delta v \leq 0.002 \text{ m/s}$ vs. Static Stop): Disabling velocity matching during conveyor grasping caused workpiece slippage and toppling, dropping conveyor grasp success from 93.3% to 36.7%.

Results of the systematic ablation study are compiled in Table XII.

F. System Latency, VRAM & Power Profiling

To verify strict adherence to the 8.0 GB VRAM budget, computational resource consumption was profiled during

TABLE XII
Systematic Ablation Study Quantifying Core Architectural
Contributions

Ablated Dimension	Configuration Tested	Task Success (%)
Kinematics Engine	TRAC-IK (Numerical)	74.0%
	ARIA Closed-Form (Ours)	89.0%
Cognitive Reasoning	Direct Single-Turn LLM	61.0%
	Linear Chain-of-Thought	78.0%
	Tree-of-Thoughts (Ours)	89.0%
Perceptual Grounding	Raw Uncalibrated Disparity	28.0%
	Claw Grounded (Ours)	89.0%
Conveyor Tracking	Static Clamping ($\Delta v > 0$)	36.7%
	Velocity Matching (Ours)	93.3%

TABLE XIII
Computational Latency and VRAM Profiling Breakdown Across
ARIA Subsystems

Subsystem / Node	Execution Frequency	Latency (ms)
Overhead Vision (YOLOv8m + SAM2)	30 Hz	18.2 ± 2.1
Wrist Depth (Depth-Anything v2)	15 Hz	28.4 ± 3.4
Local LLM Planner (Ollama 4-bit)	Event-driven	320 ± 45
Analytical IK Solver	50 Hz	0.08 ± 0.01
State Bus & ROS 2 Middleware	100 Hz	1.2 ± 0.2
FastAPI & Web Dashboard	10 Hz	8.5 ± 1.1
Complete Integrated Stack	50 Hz Control	—

continuous multi-agent execution (Table XIII). The entire multi-agent stack consumes 7,180 MB (7.18 GB) VRAM, representing 89.8% utilization on the host 8.0 GB GPU. Total system electrical power draw (laptop + arm servos + cameras) was measured at 68.5 W during peak active manipulation.

G. Failure Mode Taxonomy & Automated Recovery Analysis

Across 100 benchmark execution trials, 11 total task failures occurred (89.0% success). We classify all observed failures into five root cause categories (Table XIV):

- 1) Category A: Visual Occlusion (3 trials): End-effector arm linkage temporarily occluded overhead camera view during close-quarters placement. Recovered via wrist camera visual servoing.
- 2) Category B: Grasp Slip (3 trials): Workpiece slipped during rapid acceleration due to low friction on oily acrylic surface. Recovered via re-grasp attempt.
- 3) Category C: Conveyor Velocity Jitter (2 trials): Micro-stalls in conveyor DC motor caused minor intercept offset ($\Delta x = 4.2 \text{ mm}$).
- 4) Category D: LLM Ambiguity (2 trials): Operator prompt possessed linguistic ambiguity; resolved via HITL dialogue modal.
- 5) Category E: Servo Backlash / Thermal Drift (1 trial): Cumulative thermal drift in shoulder servo caused a 2.1 mm vertical error after 2.5 hours of continuous operation.

TABLE XIV
Systematic Failure Mode Taxonomy and Recovery Statistics (100 Experimental Trials)

Failure Category	Count	% of Failures	Automated Recovery Protocol
A: Visual Occlusion	3	27.3%	Switch to wrist vision sensors
B: Grasp Slip	3	27.3%	Increase normal contact force
C: Conveyor Speed Jitter	2	18.2%	Kalman filter velocity estimation
D: Linguistic Ambiguity	2	18.2%	Trigger Web HITL dialogue
E: Thermal Servo Drift	1	9.1%	Automated zero-offset recalibration

TABLE XV
System Deployment Capital Cost Comparison (USD)

Hardware Subsystem	Standard VLA Research Setup	Project ARIA
Robotic Manipulator	Franka Emika Panda (\$28,000)	Techno-Tactile Dual Reach (\$1,200)
Vision Sensors	Photoneo / RealSense D435i (\$1,400)	Consumer Laptop Camera (\$150)
Computing Hardware	NVIDIA A100/H100 Server (\$9,000)	Single-DoF Gripper (\$50)
End-Effector Gripper	Robotiq 2F-85 Adaptive (\$5,000)	ESP32 Microcontroller (\$5)
Tool-Change Interfacing	Industrial PLC & Drivers (\$1,500)	Software Library (\$0)
Software Licensing	Proprietary Control Suite (\$2,500)	Open-Source (\$0)
Total Capital Cost	\$47,400 USD	\$1,805 USD
Cost Reduction	Baseline	99.6%

X. Discussion, Limitations & Future Enhancements

A. Discussion of Findings: Modular Multi-Agent vs. Monolithic Foundations

The empirical findings established across our benchmarks provide critical scientific insights into the architectural design of embodied artificial intelligence. Contemporary robotics trends heavily emphasize monolithic end-to-end Vision-Language-Action policies (e.g., OpenVLA, π_0), where pixels and language tokens are directly translated into motor setpoints through massive neural networks. However, our experimental comparisons reveal that monolithic foundation policies exhibit fundamental structural vulnerabilities when deployed on resource-constrained, low-cost manipulators:

- 1) **Compounding Error Distributions:** In monolithic policies, perception errors, semantic misunderstanding, and motor inaccuracies are irrevocably entangled. If the visual encoder misinterprets a shadow, the output joint setpoints diverge uncontrollably, with no intermediate verification layer to intercept the command.
- 2) **Computational Lag and Instability:** Generating actions via autoregressive transformer tokenization incurs latencies between 85 ms and 140 ms per step. On low-inertia budget manipulators, this latency induces severe closed-loop oscillations, preventing smooth dynamic tracking of moving conveyor targets.
- 3) **The Black-Box Diagnostics Failure:** When a monolithic model fails, industrial operators cannot determine whether the failure arose from visual occlusion, kinematic limits, or planning hallucinations. In contrast, ARIA's modular multi-agent decomposition isolates intelligence into transparent, verifiable services communicating across an explainable State Bus.

B. Economic Viability & Deployment Democratization

A pivotal contribution of Project ARIA is the radical reduction of the total cost of ownership (TCO) for autonomous manipulation systems. Table XV compares the capital cost of deploying Project ARIA against typical academic and industrial foundation manipulation workcells. By substituting active RGB-D sensors with Zero-Cost-Depth and replacing datacenter multi-GPU servers

with an edge multi-agent pipeline, ARIA achieves high-dexterity manipulation at less than 1% of the hardware cost of standard research platforms (\$246 vs. \$38,400), democratizing physical robotics research for universities, small-scale enterprises, and developing nations.

C. Operational Limitations

Despite its high empirical performance, Project ARIA exhibits several operational limitations:

- **Ambient Lighting Sensitivity:** Because the Zero-Cost-Depth pipeline relies on commodity 2D RGB streams, extreme illumination drops (< 80 lux) or severe directional glare degrade monocular disparity estimation, increasing depth error from 8.2 mm to 18.4 mm.
- **Actuator Backlash and Thermal Drift:** Low-cost PWM hobbyist servomotors lack optical joint encoders and active cooling. Continuous operation exceeding 2.5 hours causes thermal accumulation in the motor windings, inducing potentiometer deadband drift of $\pm 1.2^\circ$.
- **Underactuated Wrist Degrees of Freedom:** Because the arm lacks an independent wrist yaw axis, the end-effector cannot rotate horizontally without displacing the base azimuth joint, restricting grasping access in tightly cluttered lateral corridors.

D. Future Enhancements

Future development trajectories will explore:

- **Flexible Tactile Fingertip Arrays:** Integrating thin-film piezoresistive pressure arrays onto the inner gripper pads to provide high-frequency (200 Hz) normal and shear force telemetry, enabling closed-loop slip preemption.
- **Multi-Arm Collaborative Workcells:** Extending the decentralized ROS 2 State Bus to coordinate multiple 5-DoF manipulators executing collaborative sorting along shared conveyor lines.
- **TensorRT-Edge INT8 Compilation:** Compiling the vision, segmentation, and depth models into INT8 TensorRT engines, enabling full standalone deployment on embedded 40 W platforms such as the NVIDIA Jetson Orin Nano.

XI. Conclusion

In this research, we presented the theoretical formulation, architectural design, physical implementation, and exhaustive empirical validation of Project ARIA (Autonomous Reasoning & Interaction Agent), a decentralized cognitive multi-agent Vision-Language-Action architecture for autonomous manipulation on low-cost 5-DoF robotic arms within an 8.0GB VRAM budget.

ARIA resolves the core challenges of physical autonomy on resource-constrained hardware through three key innovations:

- 1) Zero-Cost-Depth Perception: Fusing overhead YOLOv8m detection and SAM2 segmentation with wrist-mounted Depth-Anything v2 monocular depth, grounded via physical claw geometry ($Z_{\text{tips}} = 0.065\text{ m}$) to achieve sub-centimeter localization ($RMSE = 8.2\text{ mm}$) without active infrared depth sensors.
- 2) Sub-Millisecond Closed-Form Kinematics: An exact algebraic analytical inverse kinematics solver resolving the 5-DoF anthropomorphic chain in 0.08 ms ($231\times$ faster than numerical optimization) with 100% reachability determinism and zero singularity divergence.
- 3) Decentralized 15-Agent Architecture: Stratifying system intelligence into fifteen lifecycle-managed ROS 2 agents communicating asynchronously over a high-throughput State Bus, combining localized quantized LLM reasoning (Tree-of-Thoughts) with emoji-annotated telemetry and Human-in-the-Loop safety gating.

Extensive empirical evaluations across 10 standardized manipulation tasks demonstrated an overall success rate of 89.0%, substantially outperforming state-of-the-art foundation models including LeRobot ACT (72.0%), OpenVLA (68.0%), and π_0 (61.0%). In continuous moving conveyor operations ($v_{\text{belt}} = 0.05\text{ m/s}$), ARIA achieved a 93.3% dynamic sorting success rate with sub-millimeter palletizing precision ($RMSE = 1.14\text{ mm}$). By slashing complete system deployment costs by over 99%, Project ARIA proves that high-performance robotic autonomy does not require expensive research arms or datacenter compute, establishing a reproducible blueprint for accessible, explainable, and physically robust embodied artificial intelligence.

Author Contributions & Credits

Garv Arora conceived the system architecture, derived the kinematic formulations, developed the ROS 2 software stack and Gazebo simulations, conducted all experimental benchmarks, and drafted the manuscript.

Dr. Yuvaraj N provided research supervision, methodology evaluation, and reviewed the manuscript.

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